Southeast Conservation Blueprint Summary

for Oklahoma

Created 10/21/2026

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The Southeast Conservation Blueprint 2026

Southeast Conservation Adaptation Strategy

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About the Southeast Blueprint

The Southeast Conservation Blueprint is the primary product of the Southeast Conservation Adaptation Strategy (SECAS). It is a living, spatial plan to achieve the SECAS vision of a connected network of lands and waters across the Southeast and Caribbean. The Blueprint is regularly updated to incorporate new data, partner input, and information about on-the-ground conditions.

The Blueprint identifies priority areas based on a suite of natural and cultural resource indicators representing terrestrial, freshwater, and marine ecosystems. A connectivity analysis identifies corridors that link coastal and inland areas and span climate gradients.

For more information:

- Visit the Blueprint webpage
- Review the Blueprint 2026 Development Process
- View and download the Blueprint data and make maps on the Blueprint page of the SECAS Atlas

We're here to help!

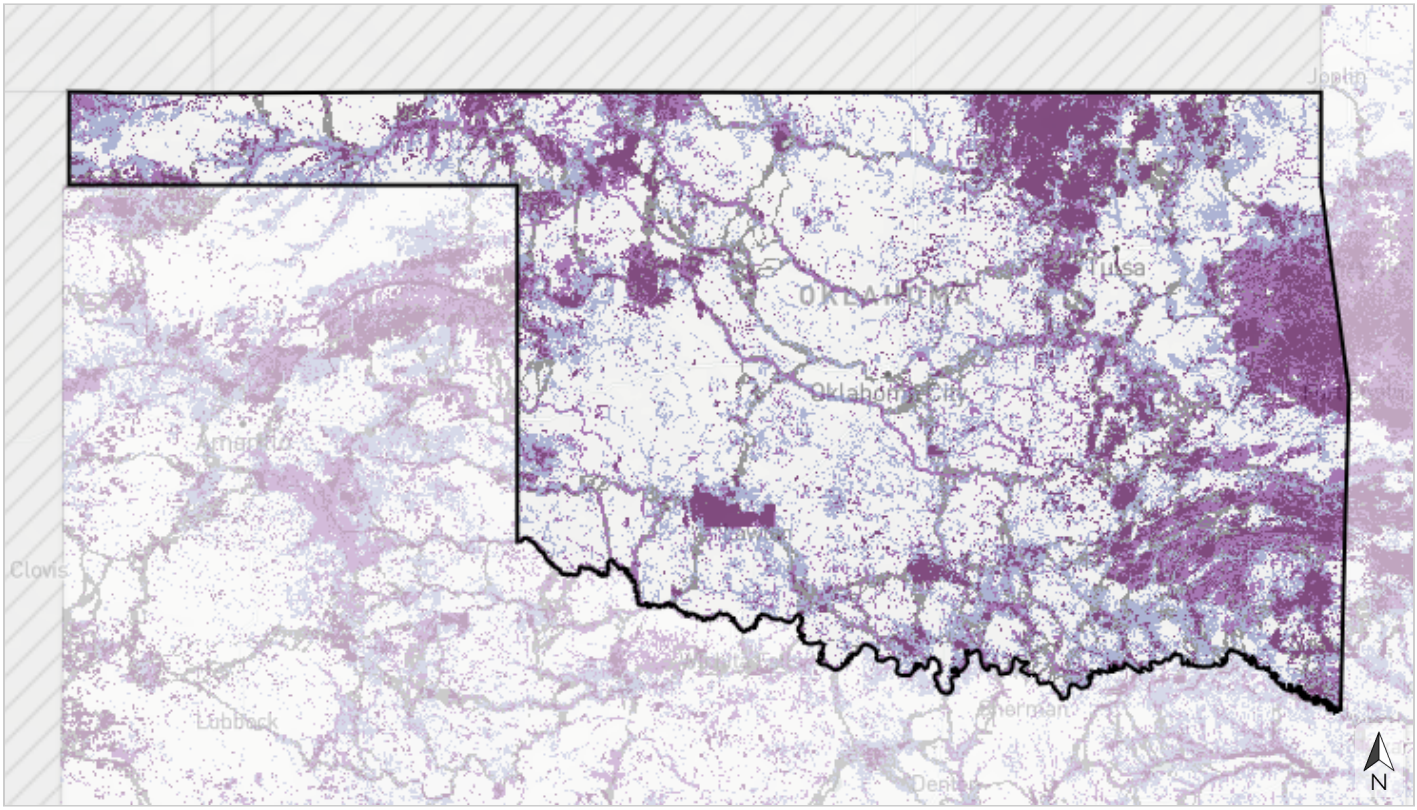
- Do you have a question about the Blueprint?
- Would you like help using the Blueprint to support a proposal or inform a decision?
- Do you have a suggestion on how to improve the Blueprint? The Blueprint and its inputs are regularly revised based on input from people like you.
- Do you have feedback on how to improve the Blueprint Explorer interface?

If you need help or have questions, contact Southeast Blueprint staff by reaching out to a member of the user support team.

We're here to support you. We really mean it. It's what we do!

This report was generated by the Southeast Conservation Blueprint Explorer, which was developed by Astute Spruce, LLC in partnership with the U.S. Fish and Wildlife Service under the Southeast Conservation Adaptation Strategy.

Southeast Blueprint Priorities



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Priorities for a connected network of lands and waters

- Highest priority
- High priority
- Medium priority
- Priority connections

Priority Categories

For a connected network of lands and waters

In total, Blueprint priorities and priority connections cover roughly 50% of the Southeast Blueprint geography.

Highest priority

Areas where conservation action would make the biggest impact, based on a suite of natural and cultural resource indicators. This class covers roughly 10% of the Southeast Blueprint geography.

High priority

Areas where conservation action would make a big impact, based on a suite of natural and cultural resource indicators. This class covers roughly 15% of the Southeast Blueprint geography.

Medium priority

Areas where conservation action would make an above-average impact, based on a suite of natural and cultural resource indicators. This class covers roughly 20% of the Southeast Blueprint geography.

Priority connections

Connections between priority areas that cover the shortest distance possible while routing through as much Blueprint priority as possible. This class covers roughly 5% of the Southeast Blueprint geography.

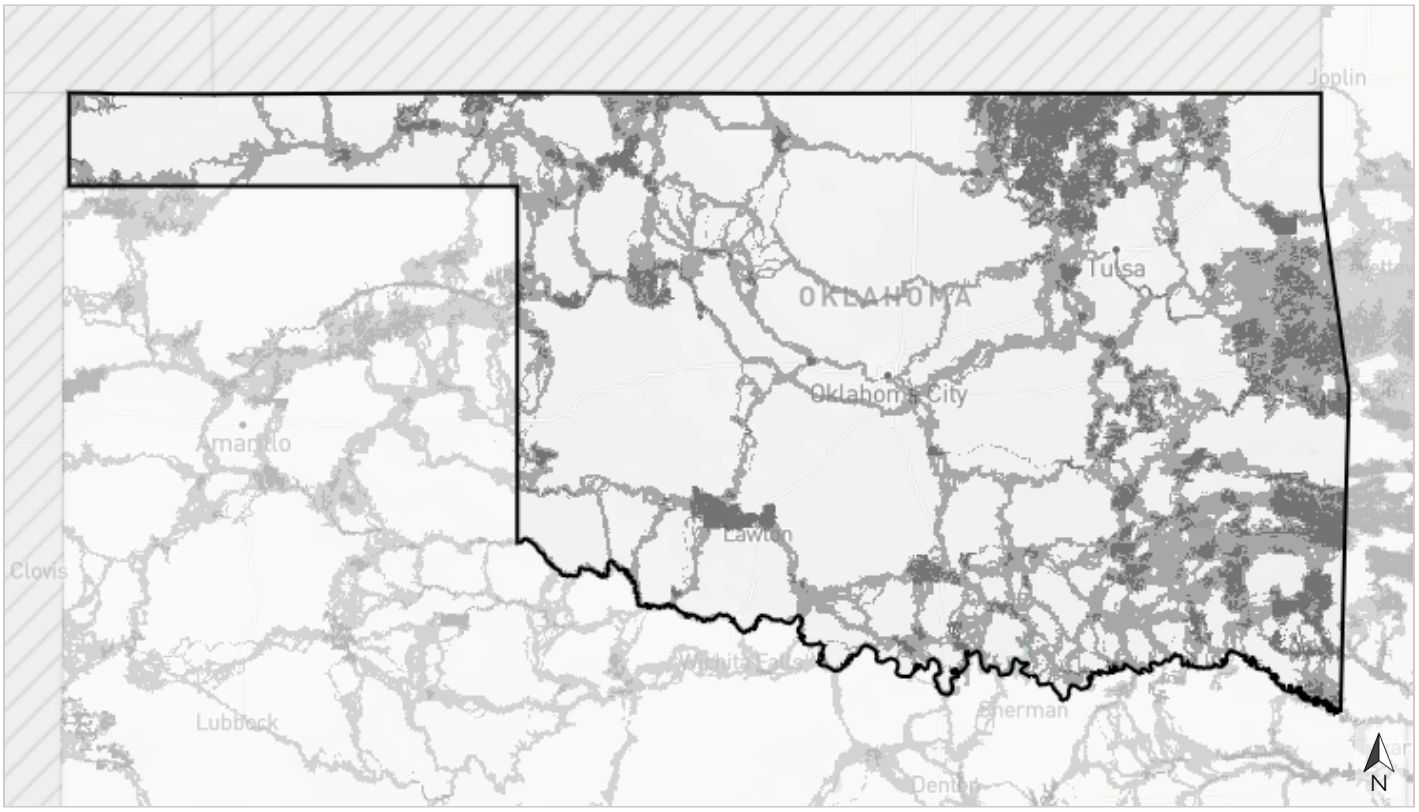
Table 1: Extent of each Blueprint priority category within Oklahoma.

Priority Category	Acres	Percent of Area
Highest priority	5,890,923	13.2%
High priority	6,223,636	13.9%
Medium priority	9,365,742	20.9%
Priority connections	2,957,430	6.6%
Lower priority	20,297,256	45.4%
Total area	**44,734,986**	**100%**

Hubs and Corridors

The Blueprint uses a least-cost path connectivity analysis to identify corridors that link hubs across the shortest distance possible, while also routing through as much Blueprint priority as possible.

In the continental Southeast, hubs are large patches (~5,000+ acres) of highest priority Blueprint areas and/or protected lands.



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Hubs
Corridors

Table 2: Extent of hubs and corridors within Oklahoma.

Type	Acres	Percent of Area
Hubs	3,881,050	8.7%
Corridors	11,201,693	25.0%
Not a hub or corridor	29,652,243	66.3%
Total area	**44,734,986**	**100%**

Indicator Summary

Table 3: Terrestrial indicators.

Indicator	Present
Fire frequency	✓
Grassland & savanna extent	✓
Grassland & savanna restoration	✓
Greenways & trails	✓
Imperiled amphibians & reptiles	✓
Imperiled mammals	✓
Intact habitat cores	✓
Landscape condition	✓
Mississippi Alluvial Valley forest birds - protection	-
Mississippi Alluvial Valley forest birds - reforestation	-
Playas	✓
Potential access to parks	✓
Resilient terrestrial sites	✓
River cane restoration	✓
Urban park size	✓
West Coastal Plain & Ouachitas forested wetland birds	✓
West Coastal Plain & Ouachitas open pine birds	✓
West Gulf Coast mottled duck nesting	-

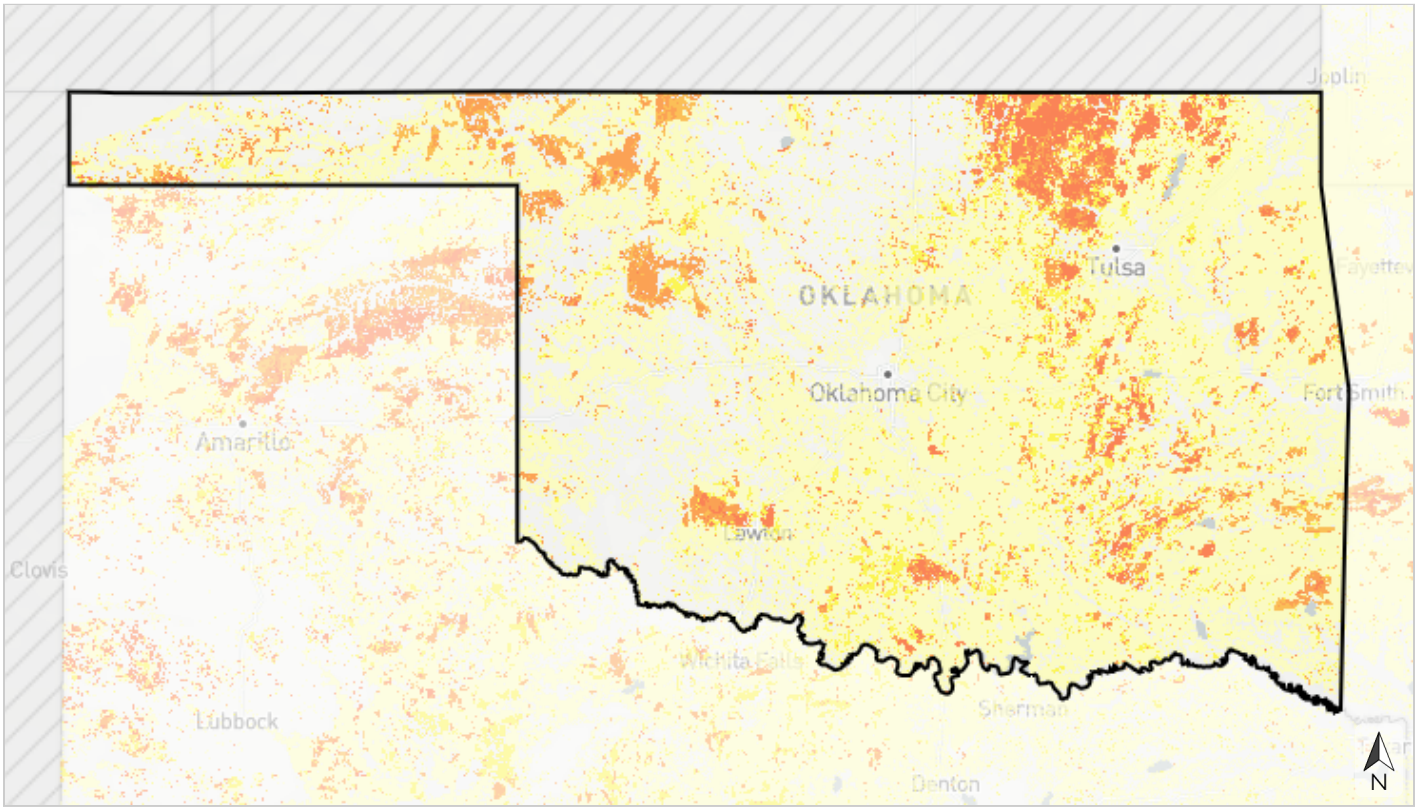
Table 4: Freshwater indicators.

Indicator	Present
Floodplain inundation	✓
Imperiled aquatic species	✓
Lakes & reservoirs	✓
Natural landcover in floodplains	✓
Network complexity	✓
Permeable surface	✓

Terrestrial

Fire frequency

This indicator uses remote sensing, fire treatment data, and historical landcover models to assess whether an area matches its historical fire regime now and in the recent past (1984-2024). It focuses on fire-adapted forests and grasslands that historically burned at least once every 20 years, which rely on regular, low-intensity fires to maintain habitat, support native plant growth, and reduce wildfire risk. It evaluates departure from historical conditions based on the U.S. Forest Service's fire regime condition classes. This indicator combines Landsat-based fire history metrics from the U.S. Geological Survey, fire treatment data from LANDFIRE, and historical fire regime estimates from LANDFIRE. Development of fire history metrics was supported by Southeast FireMap.



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How well fire-adapted areas match their historic fire regime under current conditions vs. recent past (1984-2024)

- Currently in historical fire regime, low (<33%) departure in recent past
- Currently in historical fire regime, medium (33-66%) departure in recent past
- Currently in historical fire regime, high (>66%) departure in recent past
- Not currently in historical fire regime, low (<33%) departure in recent past
- Not currently in historical fire regime, medium (33-66%) departure in recent past
- Not currently in historical fire regime, high (>66%) departure in recent past
- Urban, row crop, or not a fire-adapted area (>20-year fire regime)

Table 5: Indicator values for fire frequency within Oklahoma. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.

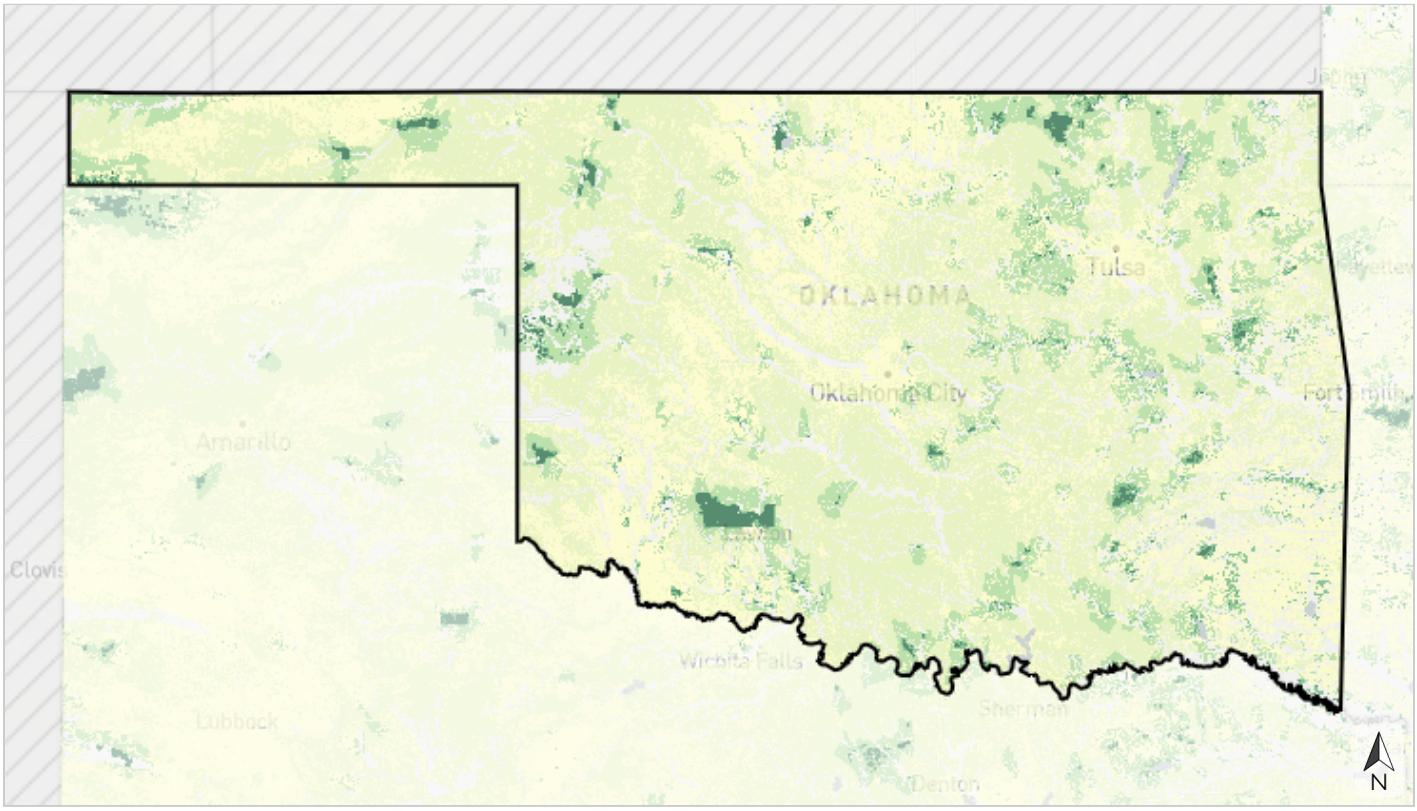
	Indicator Values: How well fire-adapted areas match their historic fire regime under current conditions vs. recent past (1984-2024)	Acres	Percent of Area	
↑ High	Currently in historical fire regime, low (<33%) departure in recent past	1,250,098	2.8%	
	Currently in historical fire regime, medium (33-66%) departure in recent past	2,513,973	5.6%	↑ In good condition
	Currently in historical fire regime, high (>66%) departure in recent past	939,530	2.1%	↓ Not in good condition
	Not currently in historical fire regime, low (<33%) departure in recent past	170,690	0.4%	
	Not currently in historical fire regime, medium (33-66%) departure in recent past	2,838,292	6.3%	
	Not currently in historical fire regime, high (>66%) departure in recent past	22,868,523	51.1%	
↓ Low	Urban, row crop, or not a fire-adapted area (>20-year fire regime)	14,153,881	31.6%	
	Total area	**44,734,986**	**100%**	

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

Grassland & savanna extent

This indicator represents grasslands and savannas in the southeastern United States, which support important plants, reptiles, amphibians, mammals, birds, and pollinators. It considers known grassland and savanna locations, predicted locations within protected areas, and surrounding pollinator buffers. It also incorporates other potential grassland and savanna locations within natural and altered landscapes, and restoration opportunities within historical locations based on past fire intervals and historic ecosystem predictions. This indicator combines data from multiple sources, including the Southeastern Grasslands Institute, the National Land Cover Database, LANDFIRE, Oklahoma and Texas ecological systems maps, and more.



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- Known grassland/savanna
- Likely grassland/savanna >10 acres
- Likely grassland/savanna ≤10 acres
- Pollinator buffer around known or likely grassland/savanna
- Potential grassland/savanna in grassland/savanna hub
- Potential grassland/savanna outside grassland/savanna hub
- Historic grassland/savanna
- Not identified as grassland/savanna

Table 6: Indicator values for grassland & savanna extent within Oklahoma. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.

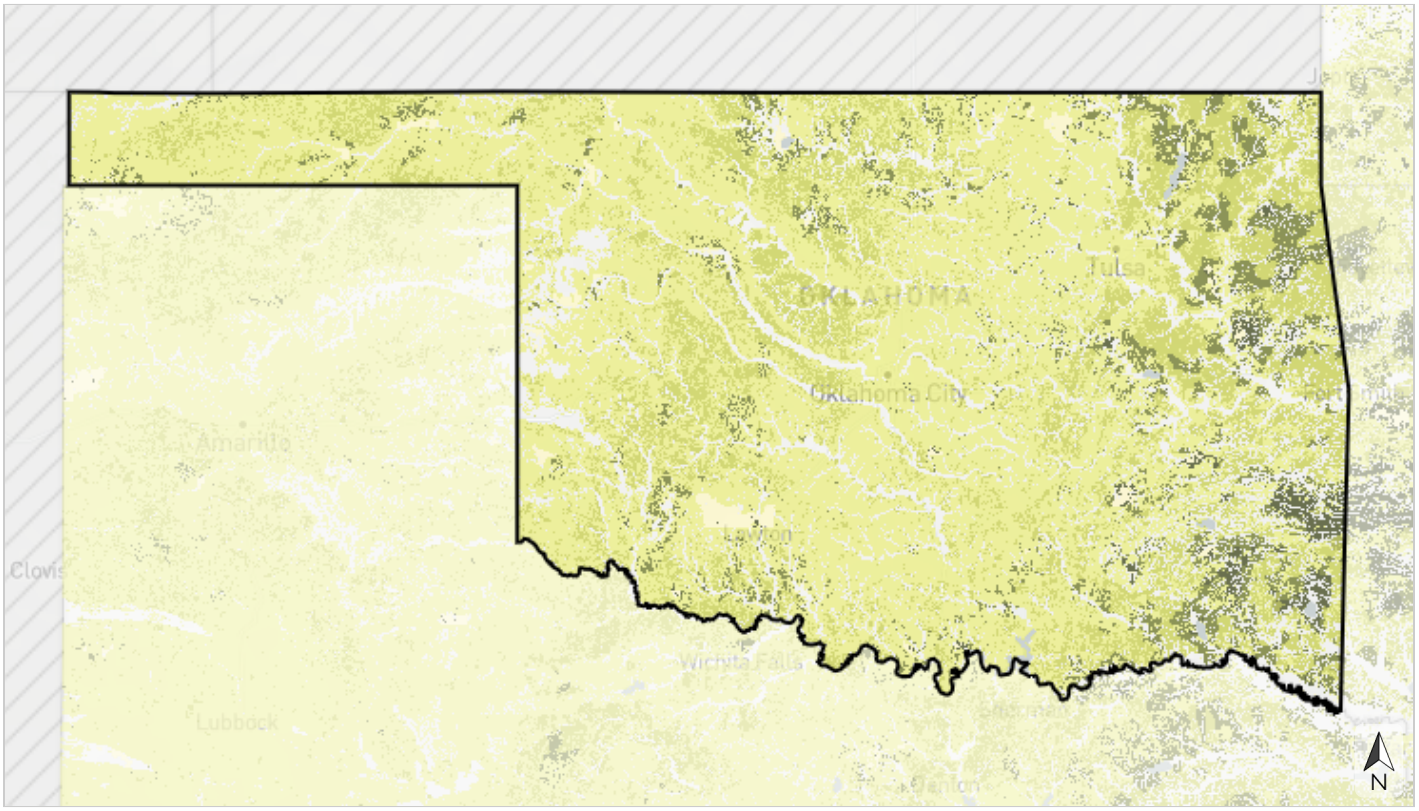
	Indicator Values	Acres	Percent of Area	
↑ High	Known grassland/savanna	0	0%	
	Likely grassland/savanna >10 acres	555,481	1.2%	
	Likely grassland/savanna ≤10 acres	32,374	<0.1%	↑ In good condition
	Pollinator buffer around known or likely grassland/savanna	652,399	1.5%	↓ Not in good condition
	Potential grassland/savanna in grassland/savanna hub	4,887,678	10.9%	
	Potential grassland/savanna outside grassland/savanna hub	15,938,495	35.6%	
	Historic grassland/savanna	14,887,345	33.3%	
↓ Low	Not identified as grassland/savanna	7,781,214	17.4%	
	Total area	**44,734,986**	**100%**	

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

Grassland & savanna restoration

This indicator represents potential restoration opportunities for grasslands and savannas within their historical range in the southeastern United States. Grasslands support important plants, reptiles, amphibians, mammals, birds, and pollinators, but have significantly declined from their current extent. It considers proximity to clusters of existing grasslands and savannas, protected areas with management potential, landcover classes with good restoration potential, priority areas for restoration of longleaf and shortleaf pine, and historical grassland and savanna locations. This indicator combines data from multiple sources, including the Southeast Blueprint grassland and savanna extent indicator, the Longleaf Suitability Analysis, shortleaf pine suitability, landcover from the National Land Cover Database and LANDFIRE, and more.



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Restoration priority

- Highest (in a grassland/savanna hub, in a protected area with management potential, in a historic patch >10 acres)
- Very high (in a grassland/savanna hub, unprotected landcover with good restoration potential, in a historic patch >10 acres)
- High (outside a grassland/savanna hub, in a protected area with management potential, in a historic patch >10 acres)
- Medium (outside a grassland/savanna hub, unprotected landcover with good restoration potential, in a historic patch >10 acres)
- Low (other historic grassland/savanna)
- Very low (already known or likely grassland/savanna)
- Lowest (not identified as historic grassland/savanna)

Table 7: Indicator values for grassland & savanna restoration within Oklahoma. A good condition threshold is not yet defined for this indicator.

	Indicator Values: Restoration priority	Acres	Percent of Area
↑ High	Highest (in a grassland/savanna hub, in a protected area with management potential, in a historic patch >10 acres)	454,347	1.0%
	Very high (in a grassland/savanna hub, unprotected landcover with good restoration potential, in a historic patch >10 acres)	2,654,248	5.9%
	High (outside a grassland/savanna hub, in a protected area with management potential, in a historic patch >10 acres)	37,159	<0.1%
	Medium (outside a grassland/savanna hub, unprotected landcover with good restoration potential, in a historic patch >10 acres)	7,970,414	17.8%
	Low (other historic grassland/savanna)	25,024,345	55.9%
	Very low (already known or likely grassland/savanna)	587,855	1.3%
↓ Low	Lowest (not identified as historic grassland/savanna)	8,006,618	17.9%
	Total area	**44,734,986**	**100%**

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

Greenways & trails

This cultural resource indicator measures both the natural condition and connected length of greenways and trails to characterize the quality of the recreational experience. Natural condition is based on the amount of impervious surface surrounding the path. Connected length captures how far a person can go without leaving a dedicated path, based on common distances for walking, running, and biking. This indicator originates from OpenStreetMap data and the National Land Cover Database.



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- Mostly natural and connected for ≥40 km
- Mostly natural and connected for 5 to <40 km or partly natural and connected for ≥40 km
- Mostly natural and connected for 1.9 to <5 km, partly natural and connected for 5 to <40 km, or developed and connected for ≥40 km
- Mostly natural and connected for <1.9 km, partly natural and connected for 1.9 to <5 km, or developed and connected for 5 to <40 km
- Partly natural and connected for <1.9 km or developed and connected for 1.9 to <5 km
- Developed and connected for <1.9 km
- Sidewalk
- Not identified as a trail, sidewalk, or other path

Table 8: Indicator values for greenways & trails within Oklahoma. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.

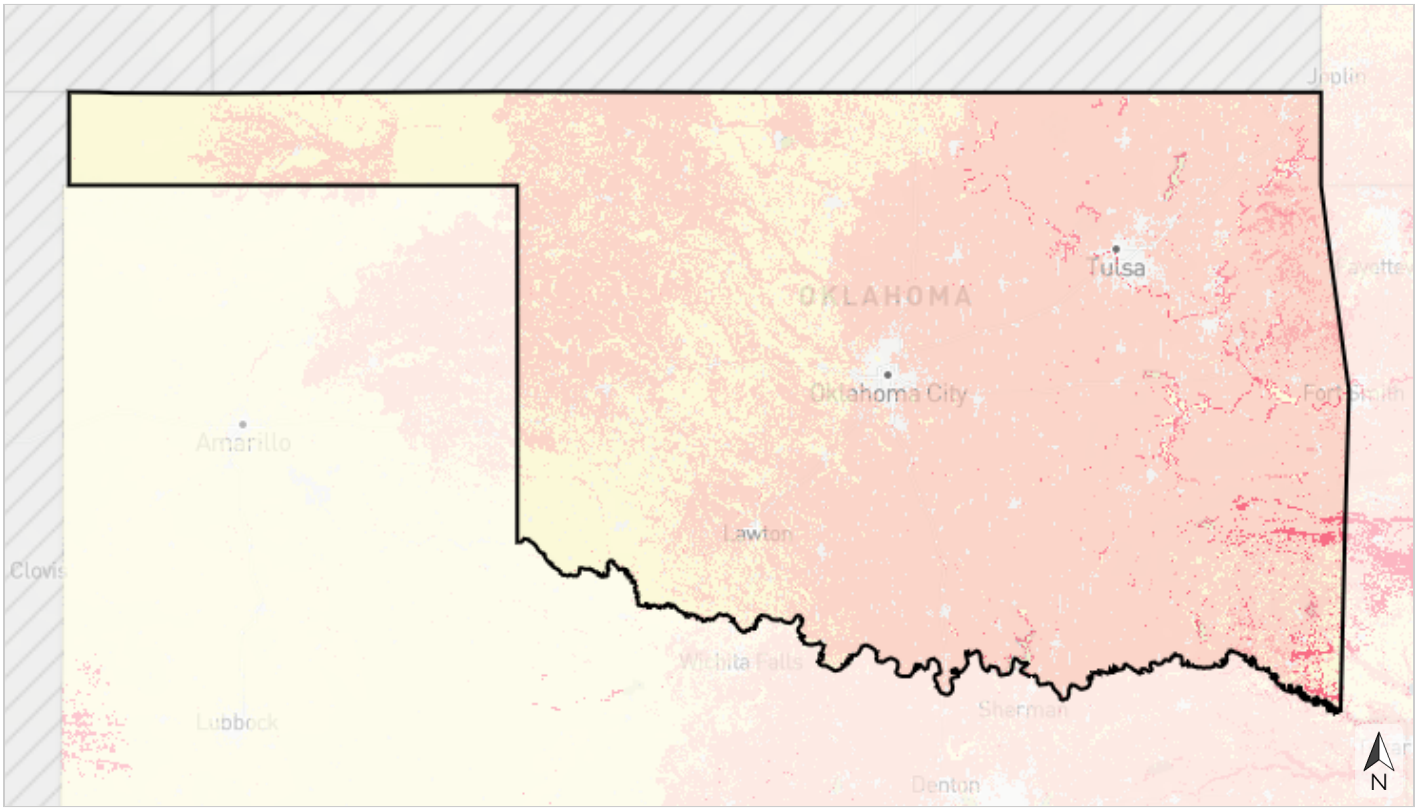
	Indicator Values	Acres	Percent of Area	
↑ High	Mostly natural and connected for ≥40 km	2,216	<0.1%	
	Mostly natural and connected for 5 to <40 km or partly natural and connected for ≥40 km	3,674	<0.1%	
	Mostly natural and connected for 1.9 to <5 km, partly natural and connected for 5 to <40 km, or developed and connected for ≥40 km	4,825	<0.1%	
	Mostly natural and connected for <1.9 km, partly natural and connected for 1.9 to <5 km, or developed and connected for 5 to <40 km	3,223	<0.1%	↑ In good condition
	Partly natural and connected for <1.9 km or developed and connected for 1.9 to <5 km	1,999	<0.1%	↓ Not in good condition
	Developed and connected for <1.9 km	3,692	<0.1%	
	Sidewalk	15,521	<0.1%	
↓ Low	Not identified as a trail, sidewalk, or other path	44,699,837	99.9%	
	Total area	**44,734,986**	**100%**	

To learn more and explore the GIS data, [view this indicator in the SECAS Atlas](#).

Terrestrial

Imperiled amphibians & reptiles

This indicator identifies potential habitat to support amphibian and reptile Regional Species of Greatest Conservation Need (RSGCN). Using a rarity-weighted richness approach, it prioritizes places important for species that are restricted to smaller ranges, such as narrow endemics, which are therefore more vulnerable to habitat loss or change. RSGCN are regional priority species derived from the list of SGCN identified in Southeast State Wildlife Action Plans as most in need of conservation action. RSGCN were chosen based on consistent criteria, such as level of conservation concern, regional stewardship responsibility, and ecological significance. This indicator uses 190 species and subspecies models from the U.S. Geological Survey Gap Analysis Project that are based on known ranges, habitat relationships, and LANDFIRE landcover.



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Importance for range-restricted species

- Highest importance (top 5%)
- Very high importance (upper 5-10%)
- High importance (upper 10-20%)
- Medium importance (above average)
- Low importance (below average)
- Developed landcover or no potential habitat

Table 9: Indicator values for imperiled amphibians & reptiles within Oklahoma. A good condition threshold is not yet defined for this indicator.

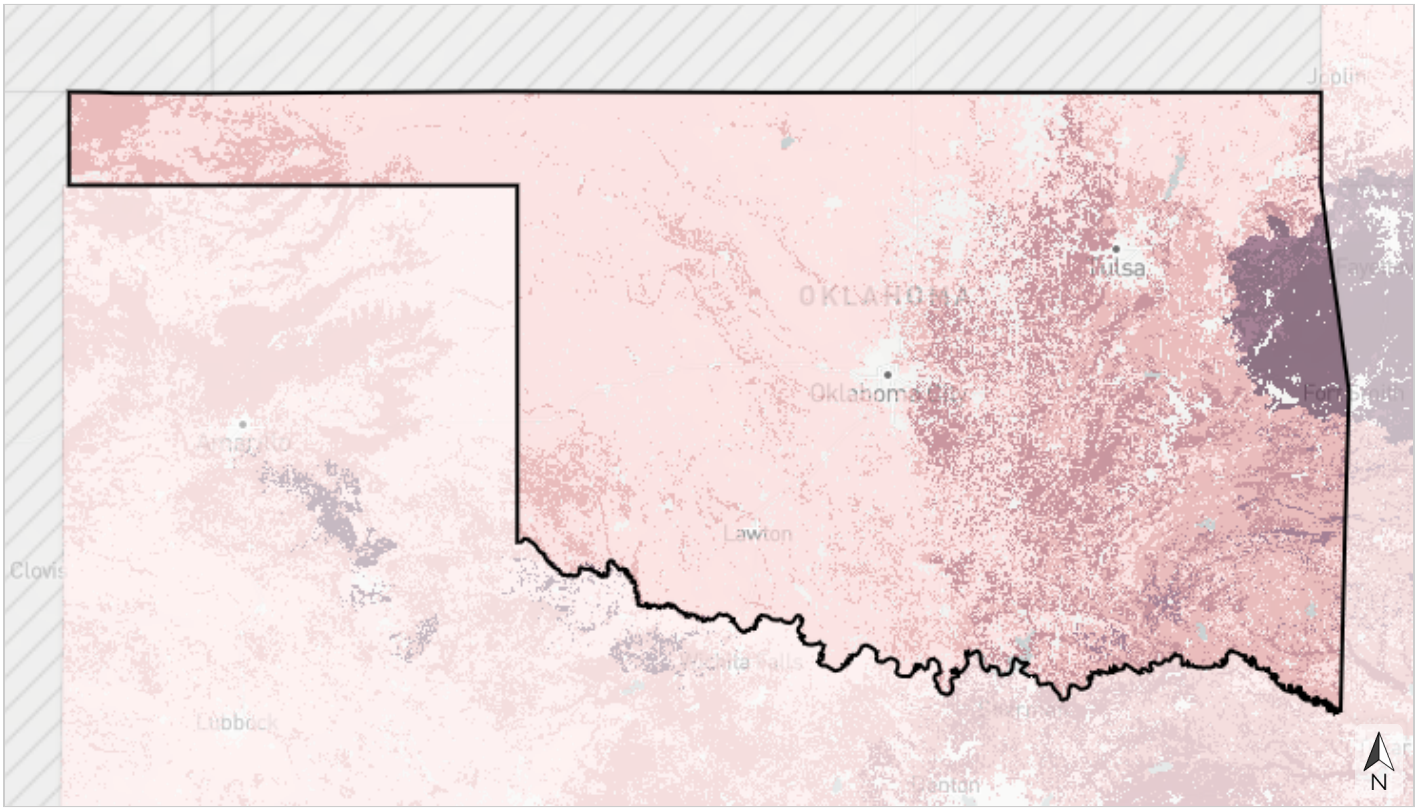
	Indicator Values: Importance for range-restricted species	Acres	Percent of Area
↑ High	Highest importance (top 5%)	624,832	1.4%
	Very high importance (upper 5-10%)	465,051	1.0%
	High importance (upper 10-20%)	1,288,803	2.9%
	Medium importance (above average)	27,642,900	61.8%
	Low importance (below average)	11,060,516	24.7%
↓ Low	Developed landcover or no potential habitat	3,652,885	8.2%
	Total area	**44,734,986**	**100%**

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

Imperiled mammals

This indicator identifies potential habitat to support mammal Regional Species of Greatest Conservation Need (RSGCN). Using a rarity-weighted richness approach, it prioritizes places important for species that are restricted to smaller ranges, such as narrow endemics, which are therefore more vulnerable to habitat loss or change. RSGCN are regional priority species derived from the list of SGCN identified in Southeast State Wildlife Action Plans as most in need of conservation action. RSGCN were chosen based on consistent criteria, such as level of conservation concern, regional stewardship responsibility, and ecological significance. This indicator uses 58 species models from the U.S. Geological Survey Gap Analysis Project that are based on known ranges, habitat relationships, and LANDFIRE landcover.



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Importance for range-restricted species

- Highest importance (top 5%)
- Very high importance (upper 5-10%)
- High importance (upper 10-20%)
- Medium importance (above average)
- Low importance (below average)
- Developed landcover or no potential habitat

Table 10: Indicator values for imperiled mammals within Oklahoma. A good condition threshold is not yet defined for this indicator.

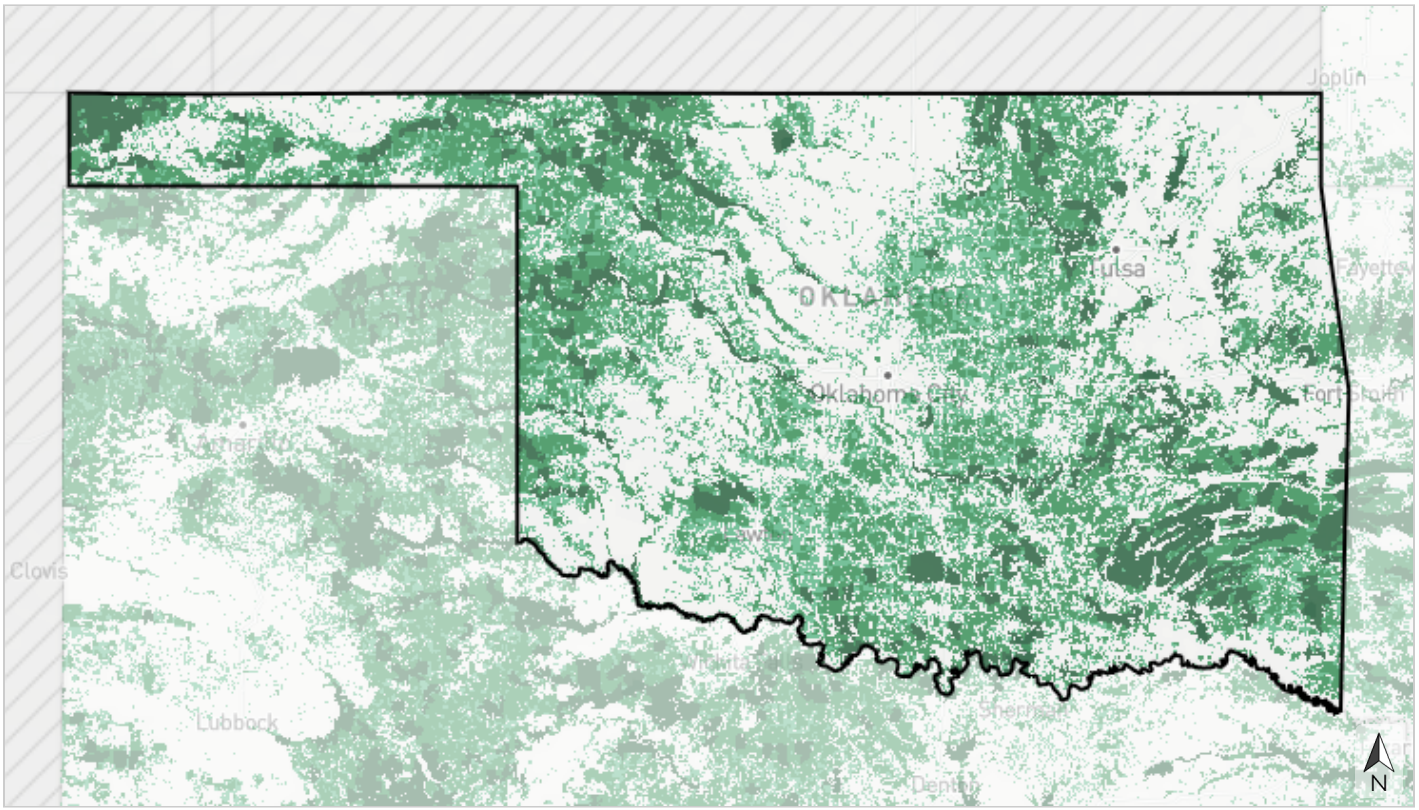
	Indicator Values: Importance for range-restricted species	Acres	Percent of Area
↑ High	Highest importance (top 5%)	1,143,479	2.6%
	Very high importance (upper 5-10%)	739,199	1.7%
	High importance (upper 10-20%)	4,229,810	9.5%
	Medium importance (above average)	9,299,948	20.8%
	Low importance (below average)	24,389,987	54.5%
↓ Low	Developed landcover or no potential habitat	4,932,565	11.0%
	Total area	**44,734,986**	**100%**

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

Intact habitat cores

This indicator represents the size of large, unfragmented patches of natural habitat. It identifies minimally disturbed natural areas at least 100 acres in size and greater than 200 meters wide. Large areas of intact natural habitat are important for many wildlife species, including reptiles and amphibians, birds, and large mammals. This indicator originates from Esri's green infrastructure data.



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- Large core (>10,000 acres)
- Medium core (>1,000-10,000 acres)
- Small core (>100-1,000 acres)
- Not a core

Table 11: Indicator values for intact habitat cores within Oklahoma. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.

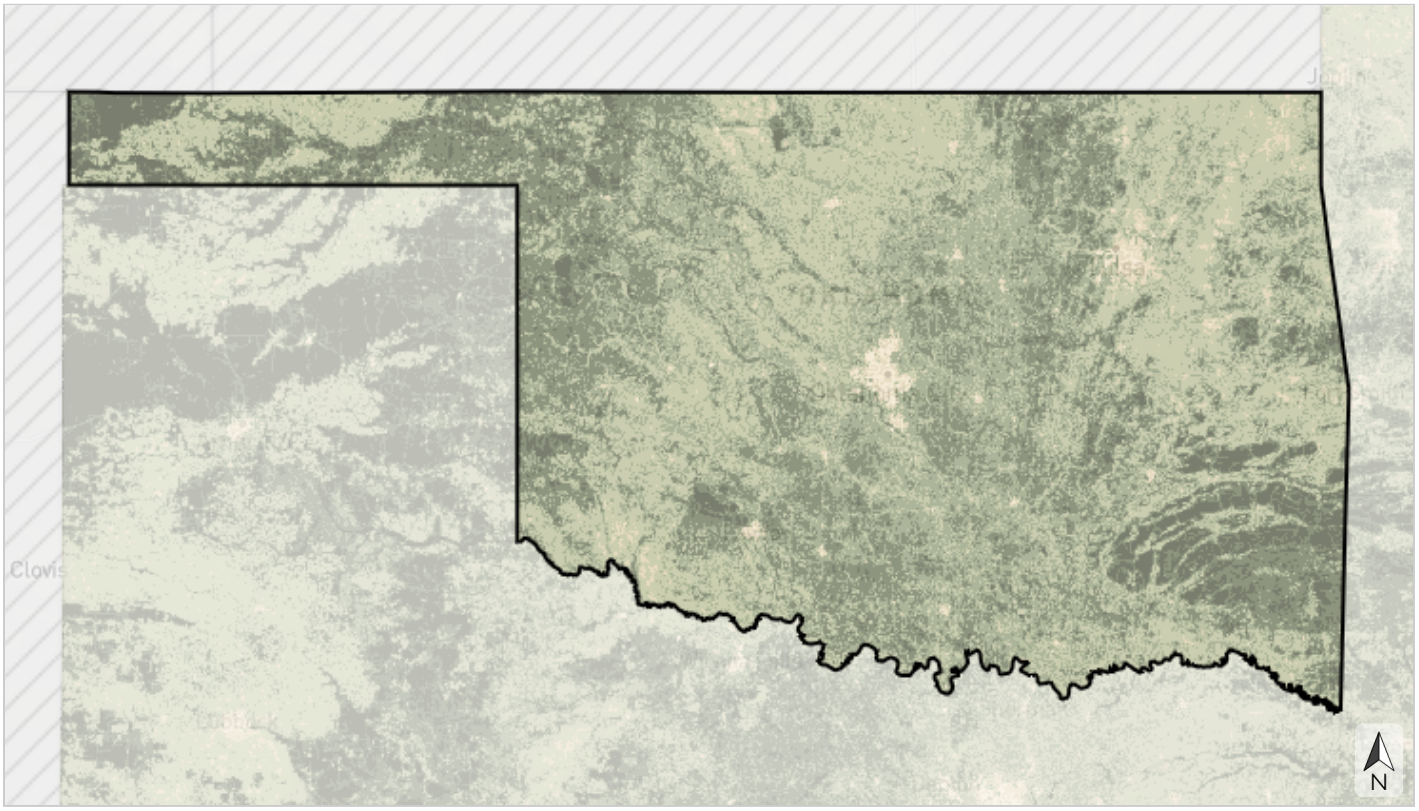
	Indicator Values	Acres	Percent of Area	
↑ High	Large core (>10,000 acres)	4,965,066	11.1%	
	Medium core (>1,000-10,000 acres)	9,852,689	22.0%	
	Small core (>100-1,000 acres)	6,995,186	15.6%	↑ In good condition
↓ Low	Not a core	22,922,045	51.2%	↓ Not in good condition
	Total area	**44,734,986**	**100%**	

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

Landscape condition

This indicator represents natural areas with limited human alteration while also considering the naturalness of the surrounding landscape. Examples of human alteration include urban development and intense agricultural use. The degree of naturalness across the landscape is a key ecological condition for sustaining species and ecosystem services that are sensitive to habitat fragmentation at multiple scales. This indicator combines data from multiple sources, including the National Land Cover Database; various data on grasslands, mines, quarries, and reservoirs; ideas from the Florida Critical Lands and Waters Identification Project's approach for evaluating land use intensity; and more.



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- Very natural landscape
- Natural landscape
- Mostly natural landscape
- Partly natural landscape
- Altered landscape
- Heavily altered landscape

Table 12: Indicator values for landscape condition within Oklahoma. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.

	Indicator Values	Acres	Percent of Area	
↑ High	Very natural landscape	2,821,961	6.3%	
	Natural landscape	11,223,455	25.1%	
	Mostly natural landscape	14,177,030	31.7%	↑ In good condition
	Partly natural landscape	15,407,829	34.4%	↓ Not in good condition
	Altered landscape	919,091	2.1%	
↓ Low	Heavily altered landscape	185,620	0.4%	
	Total area	**44,734,986**	**100%**	

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

Playas

This indicator represents the condition and location of playas, which are round, shallow depressions found primarily in the western Great Plains that serve as temporary wetlands by collecting water from rainfall and runoff. It defines a healthy playa as one that is not farmed, hydrologically modified, within a wind farm, or impacted by sediment accumulation due to agriculture. It also considers the increased benefits to wildlife provided by clusters of nearby playas, compared to more sparsely distributed playas. Playas play a critical role in recharging the Ogallala aquifer and provide habitat and food for birds and other animals. This indicator originates from the Playa Lakes Joint Venture probable playas dataset.



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- Healthy playa and part of a larger cluster
- Healthy playa
- Other playa
- Not identified as a playa

Table 13: Indicator values for playas within Oklahoma. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.

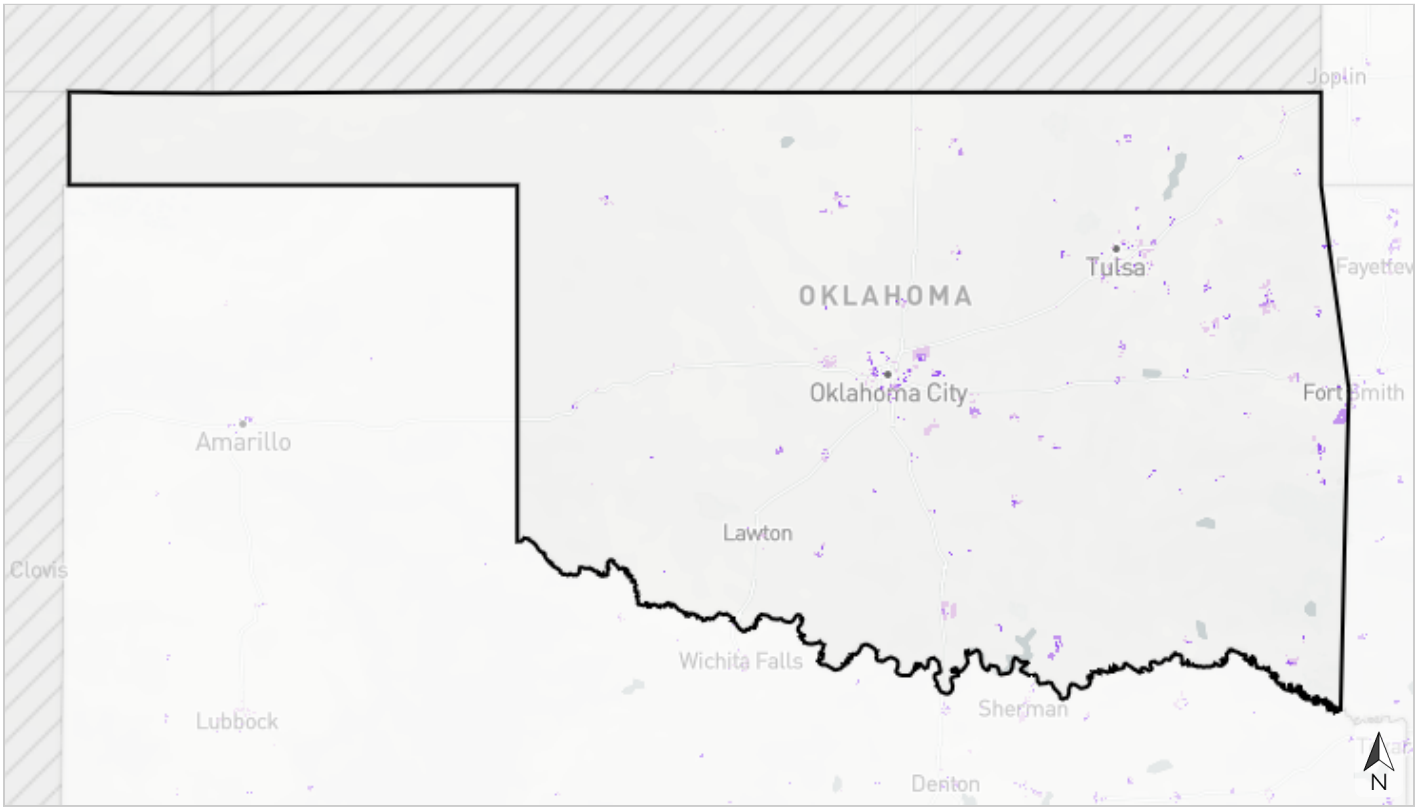
	Indicator Values	Acres	Percent of Area	
↑ High	Healthy playa and part of a larger cluster	4,752	<0.1%	
	Healthy playa	240	<0.1%	↑ In good condition
	Other playa	8,187	<0.1%	↓ Not in good condition
↓ Low	Not identified as a playa	21,287,724	47.6%	
	Area not evaluated for this indicator	*23,434,083*	*52.4%*	
	Total area	**44,734,986**	**100%**	

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

Potential access to parks

This cultural resource indicator prioritizes places to create new parks that would fill gaps in access to open space within socially vulnerable communities in urban areas. It identifies areas where residents currently lack access to parks within a 10-minute walk (accounting for walkable road networks and access barriers like highways and fences), then prioritizes based on park need using demographic and environmental metrics. Parks help improve public health, foster a conservation ethic by providing opportunities for people to connect with nature, and support critical ecosystem services. This indicator originates from the Trust for Public Land's ParkServe park priority areas and the Center for Disease Control's Social Vulnerability Index.



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Priority for a new park to serve people who lack nearby park access

- Very high priority
- High priority
- Moderate priority
- Not identified as a priority (within urban areas)

Table 14: Indicator values for potential access to parks within Oklahoma. A good condition threshold is not yet defined for this indicator.

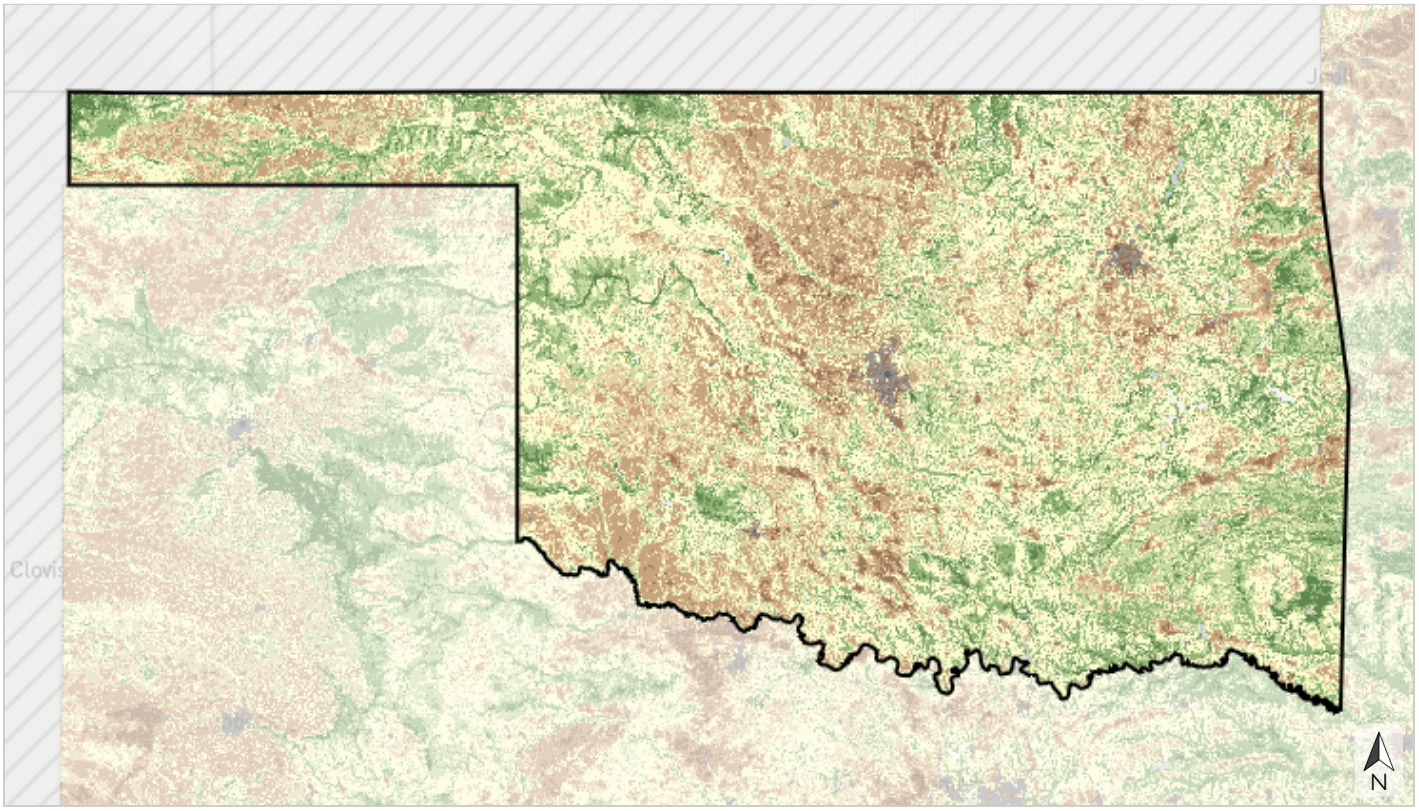
	Indicator Values: Priority for a new park to serve people who lack nearby park access	Acres	Percent of Area
↑ High	Very high priority	59,876	0.1%
	High priority	113,915	0.3%
	Moderate priority	170,890	0.4%
↓ Low	Not identified as a priority (within urban areas)	44,390,304	99.2%
	Total area	**44,734,986**	**100%**

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

Resilient terrestrial sites

This indicator depicts an area's capacity to maintain species diversity and ecosystem function in the face of climate change. It measures two factors that influence resilience. The first, landscape diversity, reflects the number of microhabitats and climatic gradients created by topography, elevation, and hydrology. The second, local connectedness, reflects the degree of habitat fragmentation and strength of barriers to species movement. Highly resilient sites contain many different habitat niches that support biodiversity, and allow species to move freely through the landscape to find suitable microclimates as the climate changes. This indicator originates from The Nature Conservancy's Resilient Land data.



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- Most resilient
- More resilient
- Slightly more resilient
- Average/median resilience
- Slightly less resilient
- Less resilient
- Least resilient
- Developed

Table 15: Indicator values for resilient terrestrial sites within Oklahoma. A good condition threshold is not yet defined for this indicator.

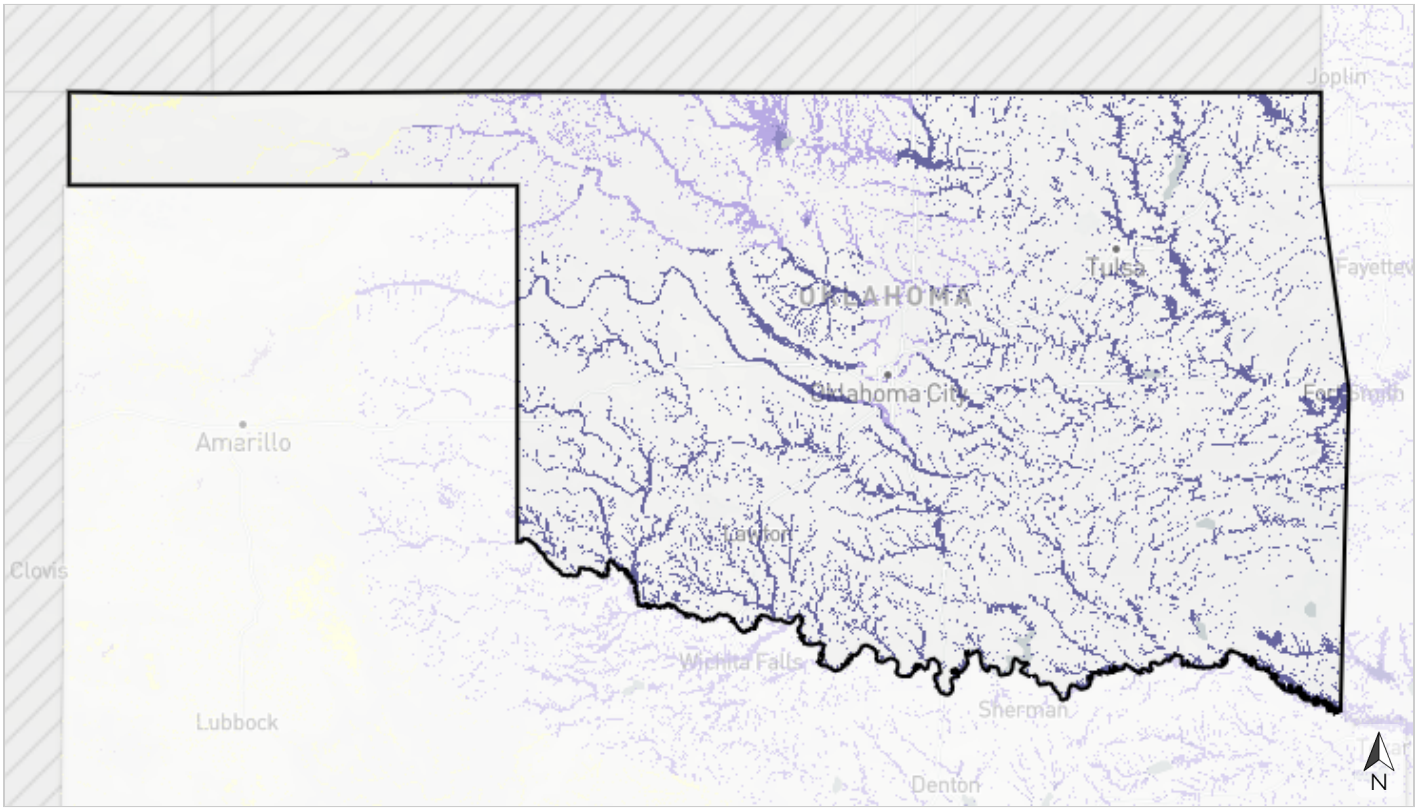
	Indicator Values	Acres	Percent of Area
↑ High	Most resilient	1,151,867	2.6%
	More resilient	6,382,134	14.3%
	Slightly more resilient	7,522,178	16.8%
	Average/median resilience	14,269,430	31.9%
	Slightly less resilient	7,130,894	15.9%
	Less resilient	6,078,374	13.6%
	Least resilient	909,926	2.0%
↓ Low	Developed	549,592	1.2%
	Area not evaluated for this indicator	*740,592*	*1.7%*
	Total area	**44,734,986**	**100%**

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

River cane restoration

This indicator represents priority places for river cane restoration and co-stewardship near the lands of federally recognized Tribes. River cane is a bamboo species native to the Southeast. Historically, it was abundant and widespread, forming dense patches called canebrakes that could stretch for miles within the floodplain. Today, it has been reduced to less than 2% of its former extent. River cane not only provides significant habitat value and ecosystem services, but many Southeastern Native American Tribes consider it a cultural keystone species important to the continuity of traditional lifeways. This indicator combines data from multiple sources, including Natural Resources Conservation Service soils data, the Environmental Protection Agency's estimated floodplain, lands of federally recognized Tribes lands from the U.S. Census Bureau, and more.



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- Potential river cane restoration/management area on Tribal land
- Potential river cane restoration/management area on protected land near Tribal lands
- Potential river cane restoration/management area near Tribal lands
- Potential river cane restoration/management area on protected land
- Potential river cane restoration/management area
- Not identified as a rivercane restoration/management area

Table 16: Indicator values for river cane restoration within Oklahoma. A good condition threshold is not yet defined for this indicator.

	Indicator Values	Acres	Percent of Area
↑ High	Potential river cane restoration/management area on Tribal land	5,864,192	13.1%
	Potential river cane restoration/management area on protected land near Tribal lands	42,697	<0.1%
	Potential river cane restoration/management area near Tribal lands	1,403,094	3.1%
	Potential river cane restoration/management area on protected land	9,946	<0.1%
	Potential river cane restoration/management area	130,691	0.3%
↓ Low	Not identified as a rivercane restoration/management area	37,284,367	83.3%
	Total area	**44,734,986**	**100%**

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

Urban park size

This cultural resource indicator measures the size of parks larger than 5 acres in the urban environment. Protected natural areas in urban environments provide urban residents a nearby place to connect with nature, and offer refugia for some species. This indicator complements the equitable access to potential parks indicator by capturing the value of existing parks. It originates from the Protected Areas Database of the United States, U.S. Census Bureau urban areas, and the National Land Cover Database.



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- 75+ acre urban park
- 50 to <75 acre urban park
- 30 to <50 acre urban park
- 10 to <30 acre urban park
- 5 to <10 acre urban park
- <5 acre urban park
- Not identified as an urban park

Table 17: Indicator values for urban park size within Oklahoma. A good condition threshold is not yet defined for this indicator.

	Indicator Values	Acres	Percent of Area
↑ High	75+ acre urban park	23,616	<0.1%
	50 to <75 acre urban park	2,094	<0.1%
	30 to <50 acre urban park	2,643	<0.1%
	10 to <30 acre urban park	4,162	<0.1%
	5 to <10 acre urban park	1,906	<0.1%
	<5 acre urban park	2,015	<0.1%
↓ Low	Not identified as an urban park	44,698,550	99.9%
	Total area	**44,734,986**	**100%**

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

West Coastal Plain & Ouachitas forested wetland birds

This indicator is an index of habitat suitability for five forested wetland bird species (Acadian flycatcher, Kentucky warbler, yellow-throated warbler, prothonotary warbler, red-shouldered hawk) within bottomland hardwood forests and riparian areas in the West Gulf Coastal Plain/Ouachitas (WGCPO) Bird Conservation Region. It uses metrics like patch size, dispersal distance, and distance to water to assess the potential for habitat to support sustainable populations of these birds. This indicator originates from the Lower Mississippi Valley Joint Venture's forested wetland decision support model for the WGCPO region.



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Habitat suitability for forested wetland bird umbrella species

- High habitat suitability (score >80)
- Medium-high habitat suitability (score >60-80)
- Medium habitat suitability (score >40-60)
- Medium-low habitat suitability (score >20-40)
- Low habitat suitability (score >0-20)
- Not suitable (score = 0)

Table 18: Indicator values for West Coastal Plain & Ouachitas forested wetland birds within Oklahoma. A good condition threshold is not yet defined for this indicator.

	Indicator Values: Habitat suitability for forested wetland bird umbrella species	Acres	Percent of Area
↑ High	High habitat suitability (score >80)	17	<0.1%
	Medium-high habitat suitability (score >60-80)	4,775	<0.1%
	Medium habitat suitability (score >40-60)	10,043	<0.1%
	Medium-low habitat suitability (score >20-40)	12,346	<0.1%
	Low habitat suitability (score >0-20)	20,671	<0.1%
↓ Low	Not suitable (score = 0)	7,286,602	16.3%
	Area not evaluated for this indicator	*37,400,532*	*83.6%*
	Total area	**44,734,986**	**100%**

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

West Coastal Plain & Ouachitas open pine birds

This indicator identifies areas with pine trees that, if managed for open condition, could support a population of three umbrella bird species (brown-headed nuthatch, Bachman's sparrow, red-cockaded woodpecker). It evaluates potential habitat in the West Gulf Coastal Plain/Ouachitas (WGCPO) Bird Conservation Region based on each species' habitat needs and population dynamics, prioritizing opportunities to restore and manage habitat to benefit open pine birds. Final scores reflect both the selectiveness of the species and whether an area meets the habitat requirements through one large patch, or clusters of smaller patches in sufficiently close proximity for breeding pairs to disperse. This indicator updates the Lower Mississippi Valley Joint Venture's open pine decision support model for the WGCPO region.



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Ability of pine patch to support a population of umbrella bird species if managed in open condition

- Large enough to support a population of all 3 species
- Large enough to support a population of 2 species
- Large enough to a population of 1 species
- Part of a cluster of nearby patches able to support a population of all 3 species
- Part of a cluster of nearby patches able to support a population of 2 species
- Part of a cluster of nearby patches able to support a population of 1 species
- Pine patch too small and isolated to support a population of any species or not an upland pine patch

Table 19: Indicator values for West Coastal Plain & Ouachitas open pine birds within Oklahoma. A good condition threshold is not yet defined for this indicator.

	Indicator Values: Ability of pine patch to support a population of umbrella bird species if managed in open condition	Acres	Percent of Area
↑ High	Large enough to support a population of all 3 species	823,726	1.8%
	Large enough to support a population of 2 species	364,909	0.8%
	Large enough to a population of 1 species	81,651	0.2%
	Part of a cluster of nearby patches able to support a population of all 3 species	56,138	0.1%
	Part of a cluster of nearby patches able to support a population of 2 species	182,466	0.4%
	Part of a cluster of nearby patches able to support a population of 1 species	11,906	<0.1%
↓ Low	Pine patch too small and isolated to support a population of any species or not an upland pine patch	5,814,463	13.0%
	Area not evaluated for this indicator	*37,399,727*	*83.6%*
	Total area	**44,734,986**	**100%**

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Freshwater

Floodplain inundation

This indicator uses the frequency of inundation to measure the hydrologic function of the floodplain. In a natural floodplain, water can move freely from mainstem rivers and lakes onto different parts of the floodplain. This cycle of frequent, but not persistent, flooding supports many aquatic species and habitats, like fish, waterfowl and swamp forests. It also provides essential ecosystem services such as nutrient retention and downstream flood control. When parts of the floodplain stay constantly wet or dry, this may signal negative impacts from structures like dikes, dams, and levees, which disconnect waterbodies from their adjacent floodplains. This indicator is based on Sentinel-2 satellite imagery and methods developed by Yvonne Allen of the U.S. Fish and Wildlife Service.



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Frequency of inundation within the floodplain

- Frequent inundation (flooded in 21-90% of days with available data)
- Regular inundation (flooded in 6-20% of days with available data)
- Occasional inundation (flooded in 2-5% of days with available data)
- Persistent inundation (flooded in 91-100% of days with available data)
- No detected inundation (flooded in 0% of days with available data)
- Not identified as a floodplain

Table 20: Indicator values for floodplain inundation within Oklahoma. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.

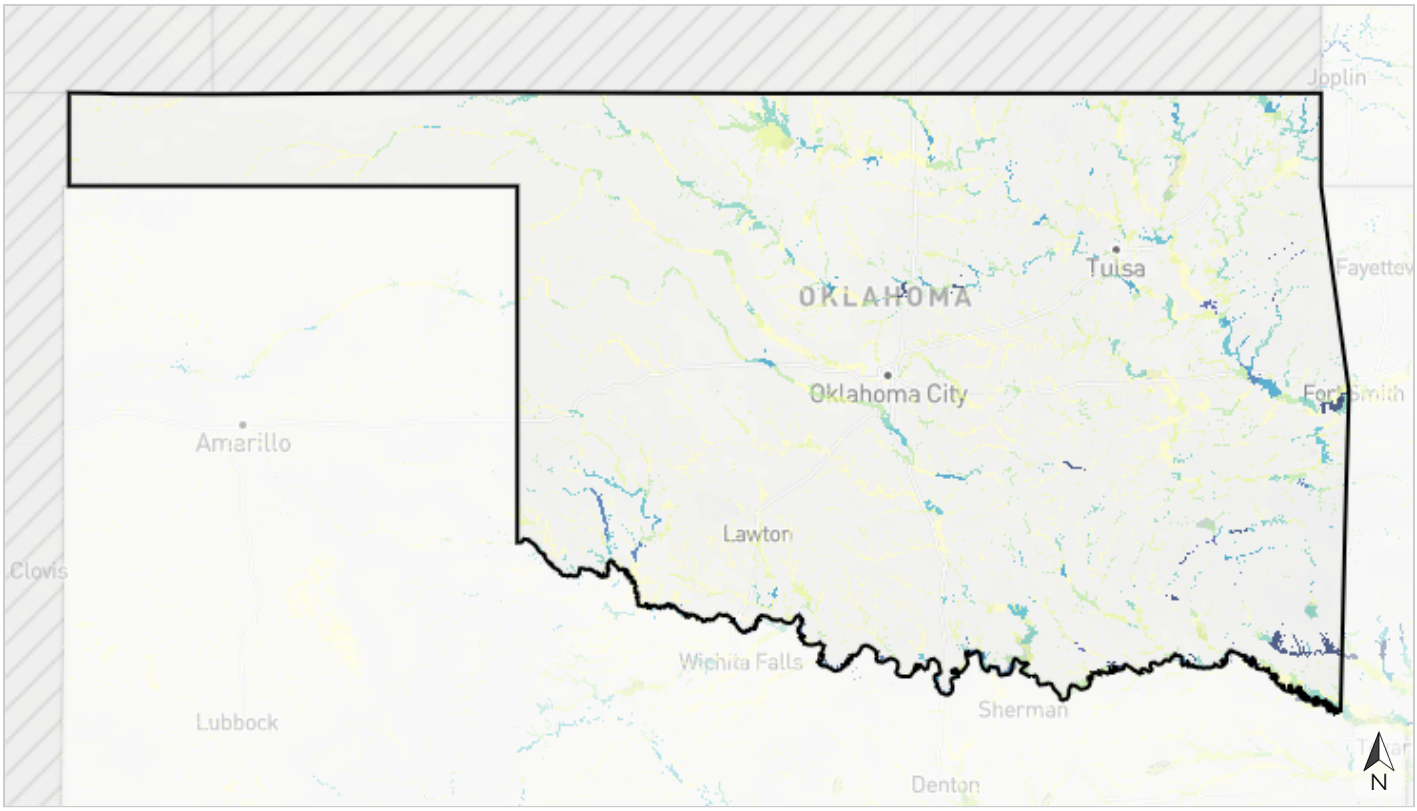
	Indicator Values: Frequency of inundation within the floodplain	Acres	Percent of Area	
↑ High	Frequent inundation (flooded in 21-90% of days with available data)	209,422	0.5%	
	Regular inundation (flooded in 6-20% of days with available data)	177,326	0.4%	↑ In good condition
	Occasional inundation (flooded in 2-5% of days with available data)	235,282	0.5%	↓ Not in good condition
	Persistent inundation (flooded in 91-100% of days with available data)	513,439	1.1%	
	No detected inundation (flooded in 0% of days with available data)	5,407,214	12.1%	
↓ Low	Not identified as a floodplain	38,192,303	85.4%	
	Total area	**44,734,986**	**100%**	

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Freshwater

Imperiled aquatic species

This indicator measures the number of aquatic animal Regional Species of Greatest Conservation Need (RSGCN) observed within each 12-digit HUC subwatershed, including fish, mussels, snails, crayfish, and amphibians. It captures patterns of rare and endemic aquatic species diversity. RSGCN are regional priority species derived from the list of SGCN identified in Southeast State Wildlife Action Plans as most in need of need of conservation action. RSGCN were chosen based on consistent criteria, such as level of conservation concern, regional stewardship responsibility, and ecological significance. This indicator originates from state Natural Heritage Program data collected by the Southeast Aquatic Resources Partnership and applies to the Environmental Protection Agency's estimated floodplain, which spatially defines areas estimated to be inundated by a 100-year flood (also known as the 1% annual chance flood).



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Number of aquatic animal Regional Species of Greatest Conservation Need (RSGCN) observed

- 8+ species
- 7 species
- 6 species
- 5 species
- 4 species
- 3 species
- 2 species
- 1 species
- 0 species
- Not identified as a floodplain

Table 21: Indicator values for imperiled aquatic species within Oklahoma. A good condition threshold is not yet defined for this indicator.

	Indicator Values: Number of aquatic animal Regional Species of Greatest Conservation Need (RSGCN) observed	Acres	Percent of Area
↑ High	8+ species	123,537	0.3%
	7 species	30,593	<0.1%
	6 species	83,225	0.2%
	5 species	164,919	0.4%
	4 species	233,324	0.5%
	3 species	538,593	1.2%
	2 species	812,001	1.8%
	1 species	1,313,854	2.9%
	0 species	3,242,637	7.2%
↓ Low	Not identified as a floodplain	38,192,303	85.4%
	Total area	**44,734,986**	**100%**

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Freshwater

Lakes & reservoirs

This indicator assesses the condition of lakes and reservoirs based on the amount of natural landcover within the upstream watershed. Higher scores go to lakes and reservoirs with less surrounding urban and agricultural development (including intensive forest management for timber production), which negatively impact drinking water quality and quantity and fish habitat. This indicator uses waterbody and watershed data from the LAGOS-LOCUS dataset, as well as landcover from the National Land Cover Database and LANDFIRE.



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- Lake or reservoir with low disturbance (0-25%) in upstream watershed
- Lake or reservoir with medium disturbance (>25-60%) in upstream watershed
- Lake or reservoir with high disturbance (>60%) in upstream watershed
- Not identified as a lake or reservoir

Table 22: Indicator values for lakes & reservoirs within Oklahoma. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.

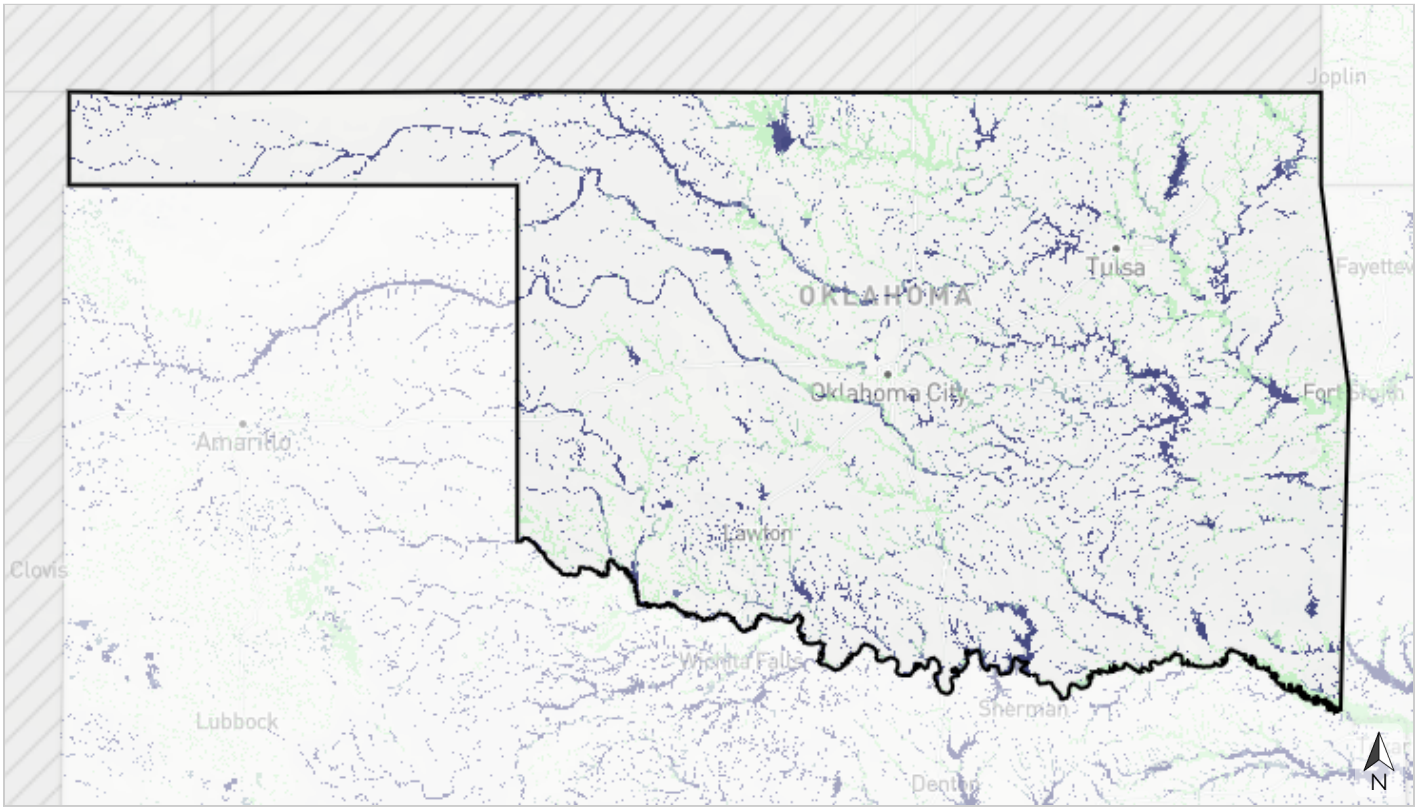
	Indicator Values	Acres	Percent of Area	
↑ High	Lake or reservoir with low disturbance (0-25%) in upstream watershed	288,237	0.6%	↑ In good condition
	Lake or reservoir with medium disturbance (>25-60%) in upstream watershed	280,891	0.6%	↓ Not in good condition
	Lake or reservoir with high disturbance (>60%) in upstream watershed	117,339	0.3%	
↓ Low	Not identified as a lake or reservoir	44,048,519	98.5%	
	Total area	**44,734,986**	**100%**	

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Freshwater

Natural landcover in floodplains

This indicator measures the amount of natural landcover in the estimated floodplain of rivers and streams within each catchment. It assesses the stream channel and its surrounding riparian buffer, measuring the percent of unaltered habitat like forests, wetlands, or open water (rather than agriculture or development). Intact vegetated buffers within the floodplain of rivers and streams provide aquatic habitat, improve water quality, reduce erosion and flooding, recharge groundwater, and more. This indicator originates from the National Land Cover Database and applies to the Environmental Protection Agency's estimated floodplain, which spatially defines areas estimated to be inundated by a 100-year flood (also known as the 1% annual chance flood).



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Percent natural landcover within the estimated floodplain, by catchment

- >90% natural landcover
- >80-90% natural landcover
- >70-80% natural landcover
- >60-70% natural landcover
- ≤60% natural landcover
- Not identified as a floodplain

Table 23: Indicator values for natural landcover in floodplains within Oklahoma. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.

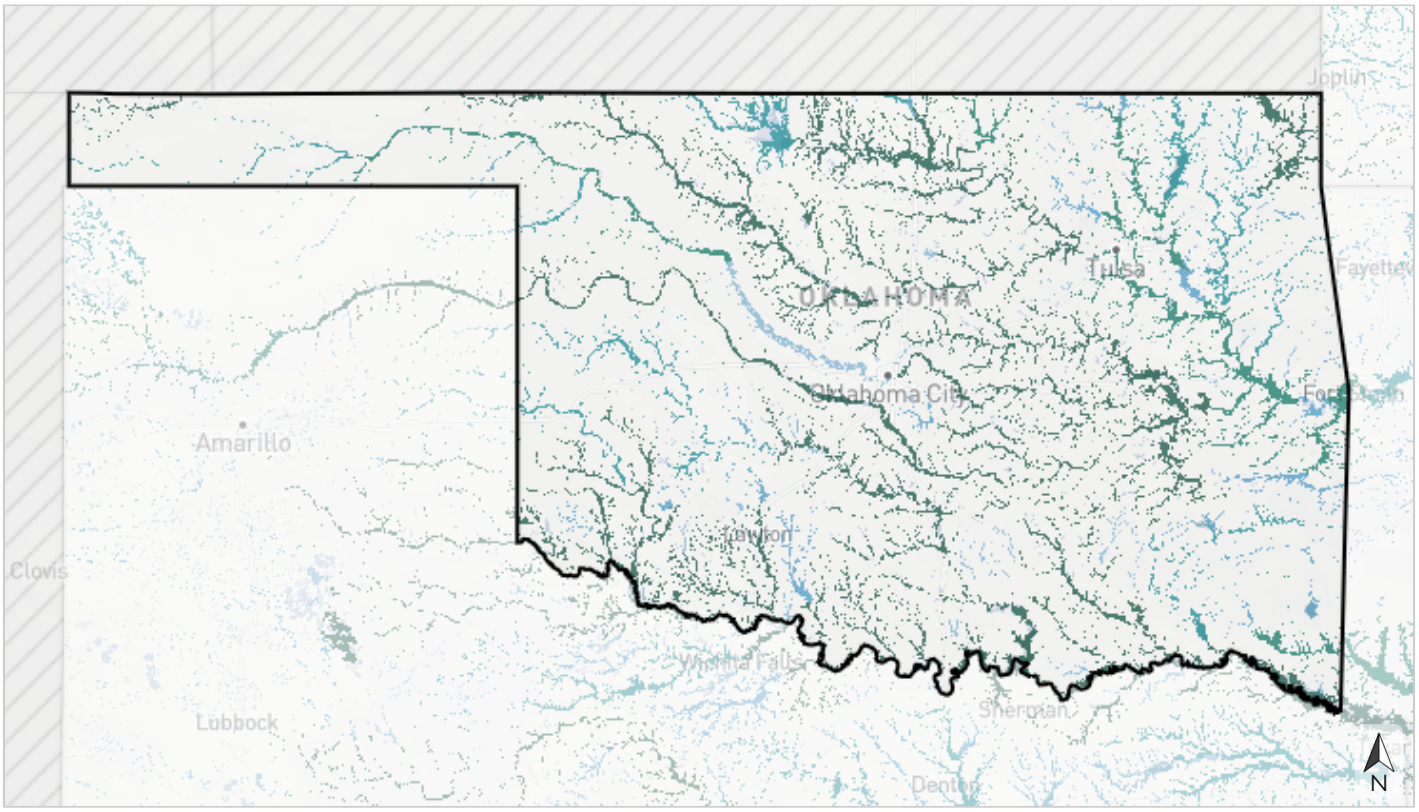
	Indicator Values: Percent natural landcover within the estimated floodplain, by catchment	Acres	Percent of Area	
↑ High	>90% natural landcover	2,235,600	5.0%	
	>80-90% natural landcover	759,380	1.7%	↑ In good condition
	>70-80% natural landcover	667,125	1.5%	↓ Not in good condition
	>60-70% natural landcover	592,283	1.3%	
	≤60% natural landcover	2,288,296	5.1%	
↓ Low	Not identified as a floodplain	38,192,303	85.4%	
	Total area	**44,734,986**	**100%**	

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Freshwater

Network complexity

This indicator depicts the number of connected stream size classes in a river network between dams or waterfalls. River networks with a variety of connected stream classes help retain aquatic biodiversity in a changing climate by allowing species to access climate refugia and move between habitats. This indicator originates from the Southeast Aquatic Resources Partnership and applies to the Environmental Protection Agency's estimated floodplain, which spatially defines areas estimated to be inundated by a 100-year flood (also known as the 1% annual chance flood).



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Number of connected stream size classes

- 7 size classes
- 6 size classes
- 5 size classes
- 4 size classes
- 3 size classes
- 2 size classes
- 1 size class
- Not identified as a floodplain

Table 24: Indicator values for network complexity within Oklahoma. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.

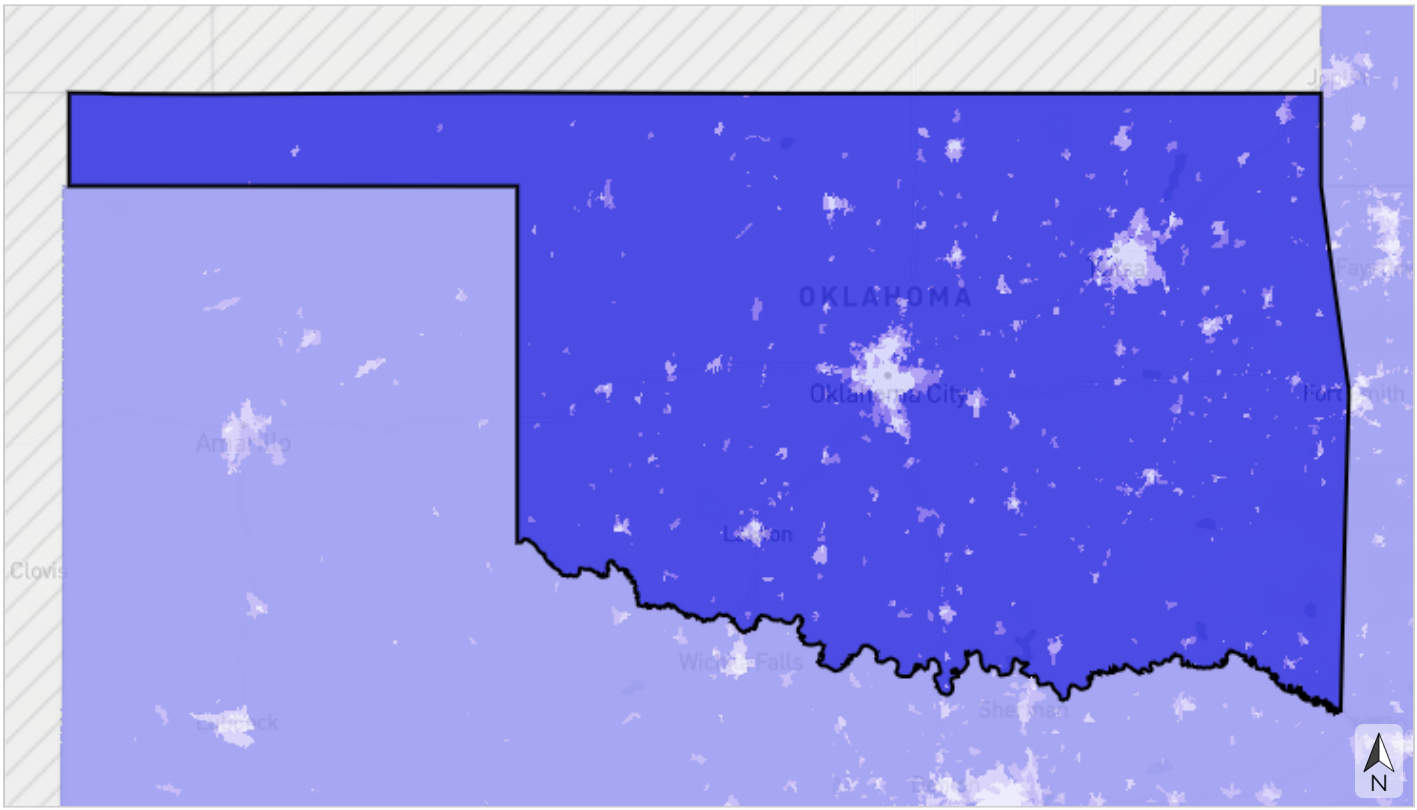
	Indicator Values: Number of connected stream size classes	Acres	Percent of Area	
↑ High	7 size classes	3,442,603	7.7%	
	6 size classes	861,312	1.9%	
	5 size classes	837,992	1.9%	
	4 size classes	418,159	0.9%	↑ In good condition
	3 size classes	274,027	0.6%	↓ Not in good condition
	2 size classes	370,084	0.8%	
	1 size class	335,570	0.8%	
↓ Low	Not identified as a floodplain	38,195,234	85.4%	
	Area not evaluated for this indicator	*4*	*<0.1%*	
	Total area	**44,734,986**	**100%**	

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Freshwater

Permeable surface

This indicator measures the average percent of non-impervious cover within each catchment. High levels of impervious surface degrade water quality and alter freshwater flow, impacting both aquatic species communities and ecosystem services for people, like the availability of clean drinking water. This indicator originates from the National Land Cover Database.



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Percent of catchment permeable

- >95% permeable (likely high water quality and supporting most sensitive aquatic species)
- >90-95% permeable (likely declining water quality and supporting most aquatic species)
- >70-90% permeable (likely degraded water quality and not supporting many aquatic species)
- ≤70% permeable (likely degraded instream flow, water quality, and aquatic species communities)

Table 25: Indicator values for permeable surface within Oklahoma. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.

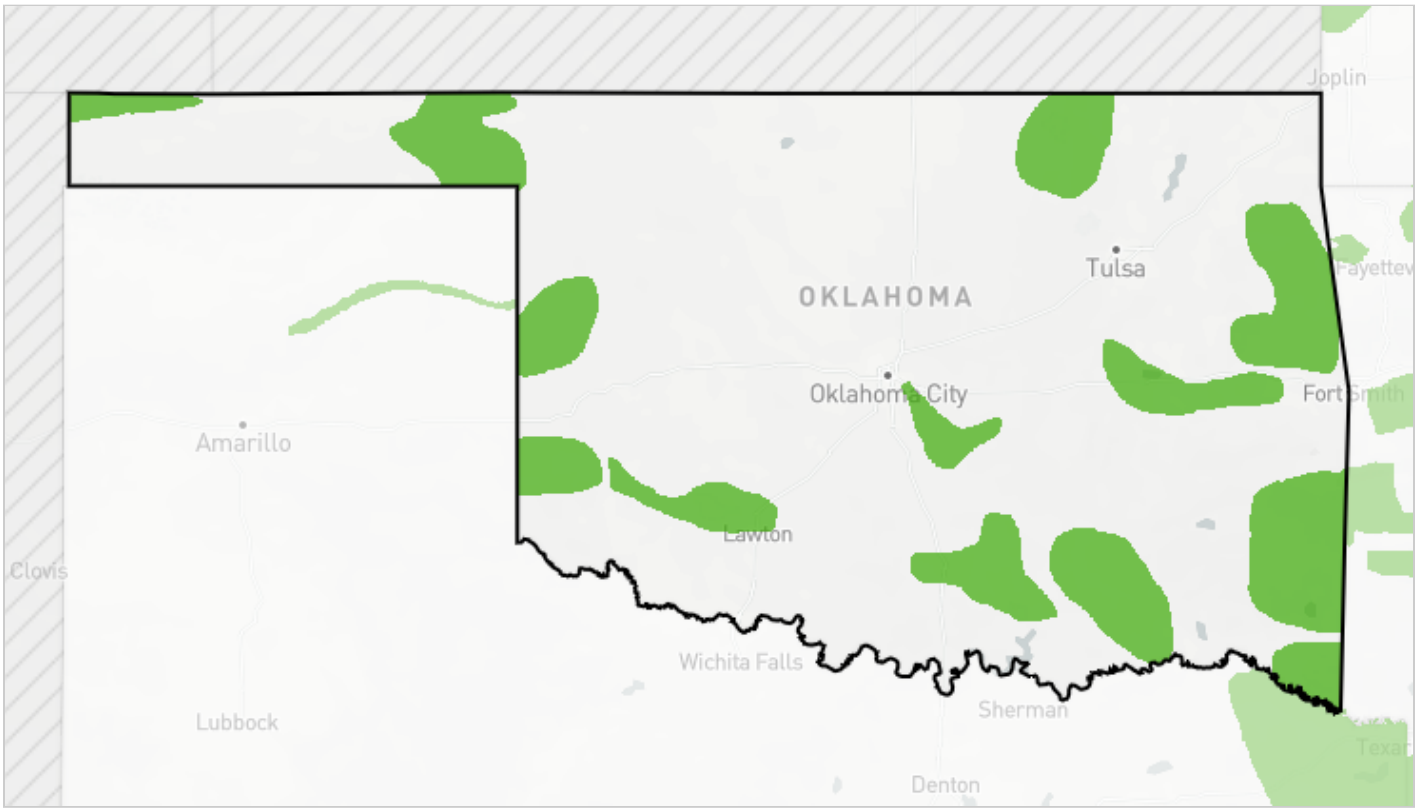
	Indicator Values: Percent of catchment permeable	Acres	Percent of Area	
↑ High	>95% permeable (likely high water quality and supporting most sensitive aquatic species)	42,694,490	95.4%	↑ In good condition
	>90-95% permeable (likely declining water quality and supporting most aquatic species)	859,641	1.9%	↓ Not in good condition
	>70-90% permeable (likely degraded water quality and not supporting many aquatic species)	801,878	1.8%	
↓ Low	≤70% permeable (likely degraded instream flow, water quality, and aquatic species communities)	378,976	0.8%	
	Total area	**44,734,986**	**100%**	

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

More Information

Priority Amphibian and Reptile Conservation Areas

Priority Amphibian and Reptile Conservation Areas (PARCAs) are an expert-driven, nonregulatory designation that includes places capable of supporting viable amphibian and reptile populations, places occupied by rare or imperiled species, and places rich in biodiversity or species unique to that geographic area. Reptiles and amphibians are a critical part of the Southeast region's rich biodiversity and many populations are declining in the face of threats like habitat loss, invasive species, and climate change. The PARCA dataset is maintained by the Amphibian and Reptile Conservancy and does not yet include Virginia or Kentucky.





Priority Amphibian and Reptile Conservation Area (PARCA)

Not a PARCA (excluding Kentucky and Virginia)

Table 26: Extent of Priority Amphibian and Reptile Conservation Areas within Oklahoma.

Priority Amphibian and Reptile Conservation Areas status	Acres	Percent of Area
Priority Amphibian and Reptile Conservation Area (PARCA)	8,586,217	19.2%
Not a PARCA (excluding Kentucky and Virginia)	36,148,770	80.8%
Total area	**44,734,986**	**100%**

Priority Amphibian and Reptile Conservation Areas at this location:

Arbuckle

The Arbuckle Mountains are some of the oldest mountains in the United States and contain one of the last free-flowing rivers in Oklahoma. This outdoor paradise includes the Arbuckle Plains, which are covered with springs, caves, and unique flora and fauna. A variety of plant communities can be found here. Cross timber communities containing blackjack and post oak woodlands and prairies combine to form a patchy mosaic, all of which are shaped by fire. Limestone outcrops provide essential habitat for snakes and lizards, including the Texas horned lizard. Lastly, bottomland hardwood forests can be found along the streams and are dominated by elms, sugarberry, and green ash. However, excessive groundwater removal, damming of rivers, and fire suppression are just a few of the threats that need to be addressed to keep this PARCA beautiful.

Beaver Grasslands

This PARCA includes part of the Shortgrass Prairie Region, which is home to a diverse range of herbaceous species. Much shortgrass prairie has been converted to other land uses, such as crop fields, ranching, housing developments, and roads. Working together with partners and landowners to convert old crop fields back to shortgrass prairies using native plants and removing exotic invasive species like Russian thistle and old world bluestem is a necessary step to mending this fragmented ecosystem.

Black Mesa

This PARCA encompasses the Black Mesa area within the Shortgrass Prairie ecoregion and is dominated by shortgrass habitat. Grama grass species, muhly grass, prickly pear cactus, and yucca are common species in this habitat. Many springs and seeps can be found in this area and eastern cottonwood-willow woodlands are prominent in riparian areas. Many reptile and amphibian species found in this area are declining due to habitat loss and fragmentation.

Boggy Bottoms

The highest herpetofaunal diversity in the state of Oklahoma can be found in Boggy Bottoms, and more than 60 species were recently identified. Naturally occurring beaver-formed wetlands surrounded by intact hardwood forest represent biodiversity hotspots. Habitats in this ecoregion may contain more than 30 plant species per square meter (a density on par with tropical forests), making it one of the most biologically rich natural systems on Earth. Rivers and streams are essential to the forested bottoms that characterize this area. During spring floods, these rivers and streams deposit soil, seeds from upstream forests, and sediments that become natural fertilizer for plants in the floodplain. These rich soils promote an abundance of floral diversity, which in turn helps to support a diverse vertebrate community.

Caddoan

The Caddoan PARCA in northeast Texas represents a variety of habitats; the eastern portion is generally forested with shortleaf pine, oaks and hickories, transitioning into post oak savanna and prairie in the western portions. Historically, the prairies contained a diverse microtopography of "mima" mounds and depressions. Much of this has been lost as they were converted to agriculture, but some examples on conservation and managed private lands remain. River floodplains contain some of the best bottomland hardwood forest habitat in the state of Texas. Species such as the pygmy rattlesnake, crawfish frog, and the Gulf Coast waterdog call this PARCA home, and they’re threatened by fire suppression, forest

conversion and clearing, prairie succession, mining, and invasive species.

Illinois River Woodland and Prairie
The Illinois River Woodland and Prairie PARCA covers a portion of Ozark National Forest and contains more than 30 amphibian species. This region is characterized by low plateaus with extensive plains that are underlain with chert and limestone. Caves and sinkholes are common in the area, along with perennial spring-fed streams. The lower elevation and widespread plains have made this area prone to development and agriculture.

Lower Arkansas River
The natural state of the lower Arkansas River was forever altered with the McClellan-Kerr Arkansas River Navigation System. Eighteen locks and dams modify the river, creating a major reservoir and access for commercial barge traffic. In addition, much of this area's tallgrass prairies and canebreak habitat have been converted to agricultural use for crops and livestock grazing, further fragmenting the land and altering ecosystems. Conservation efforts for this PARCA will include restoring native plant species and land connectivity.

Lower Little River
This important area for reptile and amphibian diversity has an intricate system of large streams, oxbow lakes, and backswamps. Covering hilly dissected uplands, floodplains, and low terraces are various species of oak, pine and cypress trees. This PARCA includes Little River National Wildlife Refuge, which is home to 11 state Champion Trees (largest of their respective species in the state) and offers refuge to several at-risk turtle species.

Ouachita Mountains East
This mountain range consists of folded ridges, rolling hills, and broad valleys formed by the erosion of Paleozoic sandstone and shale. Their east-to-west-oriented ridges make them an unusual landmark in North America, but even more impressive is the diversity of salamanders that reside within the forests. Some of the salamanders are found nowhere else and include the Rich Mountain salamander and the Fourche Mountain salamander.

Ouachita Mountains West
Open hills and low mountains (300-400 ft) with valleys filled with fissured cliffs characterize this region. The major habitat types are native mixed shortleaf pine-deciduous forests with diverse bottomland forests along streams (willow, white, and Shumard oaks, as well as black hickory, sweet gum, green ash, and maples) filled with cool, clear, spring-fed water. Large areas have experienced habitat loss with commercial silviculture planted with loblolly pine, while grasslands have been largely converted to exotic species for grazing. This area hosts the highest species richness in the state, where many species from the western and eastern states converge, and is the western extent of the range for several threatened species such as the alligator snapping turtle. Conservation efforts here will focus on preserving fishless ponds for amphibian breeding, restoring pine ecosystems with a native fire regime, and protecting stream headwater habitats.

Ozarks
The Ozark Mountains are known for their many distinctive geological landscapes, including the impressive

karst region. This unique region is characterized by an underground wilderness made up of soluble limestone and dolomite. Karst features include caves, clear streams, and sinkholes, and their formation occurs slowly over time. This makes them very susceptible to pollution and vandalism. This subterranean ecosystem provides significant habitat to many species and some of these species are found nowhere else in the world, such as the grotto salamander. Other important habitats of the Ozarks are gravel-bottom streams and accompanying riparian forests, which are essential to many amphibians and reptiles. Restoring these stream and riparian habitats is a conservation goal for this PARCA, along with protecting vulnerable caves and their delicate ecosystems.

Sandy Sanders
This unique PARCA is a rolling grassland located in a less populated area of Oklahoma and in one of the warmest and driest parts of the state. Most of the land in this region is privately owned ranchland, except for a 30,000-acre state Wildlife Management Area. Mixed-grass prairie and prairie shrublands with an overstory of honey mesquite are found throughout this region. Redberry juniper woodlands and gypsum canyonland habitats with rare plant species are found on hills and canyons peppered throughout the area. Fire suppression and poor range management have encouraged brush encroachment within the prairie, leading to habitat degradation. Rattlesnake roundups threaten local populations of rattlesnakes such as the western diamondback. These top predators play a major role in their ecosystems and already face many other threats, including habitat loss, road mortality, and poaching.

Shinnery Oaks
The Shinnery Oaks PARCA is a unique landscape comprised of eroded canyonlands, old sand dunes dominated by shinnery oak, mixed-grass prairies, and riparian woodlands along perennial and intermittent streams. Historically, these shrublands were once estimated to cover about 750,000 acres of Oklahoma, but now only 116,000 acres remain. The mixed-grass prairies in this area are formed by a diverse plant composition, but little bluestem and sideoats grama are the dominant grass species. Most of this important habitat has been converted to crops and ranchland, and heavy grazing coupled with fire suppression has led to an invasion of introduced grasses and forbs. Fragmentation caused by roads is another common issue in this area, and wildlife road mortalities can be detrimental to struggling species. Figuring out a solution to minimizing road fatalities, especially with hard-to-see reptiles and amphibians, will be an objective for this PARCA.

Tallgrass Prairie
This PARCA contains the Osage Hills and a portion of the largest remaining remnants of historic tallgrass prairie. It is dominated by native warm-season prairie plants that were once widespread across 14 states. However, much of this habitat has been greatly reduced by urban sprawl and conversion to agriculture. Dominant plant species found here are bluestem, switchgrass, rattlesnake master, and Oklahoma beardtongue. Historically, prairies were shaped by wildfires and bison, but fire suppression has allowed unwanted woody encroachment. Restoring a natural fire regime, using prescribed grazing, and invasive species removal are a few ways keep the tallgrass prairies healthy and the wildlife abundant.

Texas Canadian River
True to its name, the Canadian River PARCA contains the Canadian River from below the Lake Meredith Dam all the way to the Oklahoma border, as well as adjacent riparian zone and floodplain oxbow lakes.

The riparian forests are dominated largely by cottonwoods and hackberries with some smaller tributaries containing mesquite-juniper brush. Beyond the riparian zone, there are large bands of mesquite shrub-grassland. Species such as the smooth softshell, Texas horned lizard, and plains hognose snake occupy this PARCA and are threatened by invasive plants and increased agricultural and urban water uses.

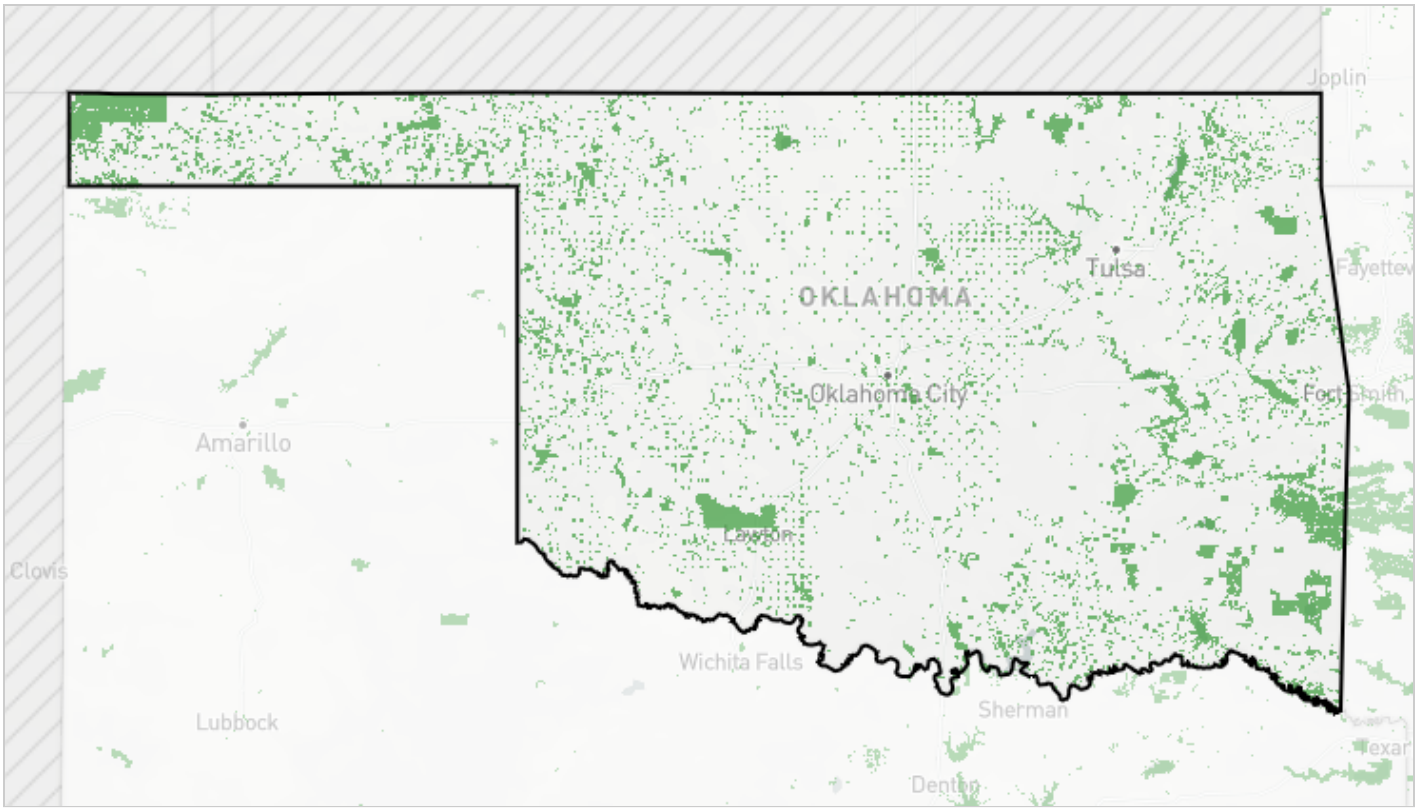
Thunderbird
Situated near Oklahoma City, the Thunderbird PARCA is a mosaic of open oak woodlands, grasslands, and riparian ecosystems. Periodic fire and drought help maintain the structure of the plant communities found within this PARCA. Dominant species include blackjack oak, post oak, sumac, hackberry, and bluestem grasses. Sandy bottom rivers like the Little River and Canadian River bisect this PARCA. Since this PARCA is surrounded by development, it is at risk of pesticide/herbicide runoff into sensitive wetlands, invasive species like fire ants, feral hogs and cats, road mortalities stemming from fragmented habitat, and tourism.

Wichita Mountains
This ancient granite mountain range is home to one of the earliest wildlife refuges found in the United States. Today, this refuge is known as Wichita Mountains National Wildlife Refuge. As the largest refuge in Oklahoma, it protects 60,000 acres of mixed-grass prairies, freshwater lakes and streams, and diverse mountain habitats. The Wichita Mountains were also the site for one of the last gold rushes, and evidence of this historical era can still be found throughout the land. Major habitats include cross timbers of blackjack and post oak, open scrubby xeric forests, and grasslands with short grasses, prickly pear, and barren rocky areas. This multipurpose area is managed for many different uses, such as tourism, wildlife protection, rangeland, farming, and energy development.

Names and descriptions of public Priority Amphibian and Reptile Areas were provided by the Amphibian and Reptile Conservancy on August 30, 2024 and edited slightly for clarity and consistency.

Protected Areas

Protected areas include a variety of public lands owned or managed by federal, state, and local agencies and nonprofit organizations, as well as some private conservation lands and conservation easements. Protected areas are derived from the Protected Areas Database of the United States (PAD-US v4.1) and include Fee, Designation, Easement, Marine, and Proclamation (Dept. of Defense lands only) boundaries.



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Within a protected area
Not within a protected area

Table 27: Extent of protected areas within Oklahoma.

Protected area status	Acres	Percent of Area
Not within a protected area	39,518,845	88.3%
Within a protected area	5,216,142	11.7%
Total area	**44,734,986**	**100%**

Protected areas at this location:

- CLO Lands (The Commissioners of the Land Office; 3,238,932 acres)
- Ouachita National Forest (USDA Forest Service; 353,440 acres)
- Eufaula Lake (101,458 acres)
- Fort Sill (93,725 acres)
- Honobia Creek Wildlife Management Area (Department of Wildlife Conservation; 80,896 acres)
- Robert S. Kerr Lake (62,676 acres)
- Wichita Mountains Wildlife Refuge (US Fish and Wildlife Service; 59,029 acres)
- McAlester Army Ammunition Plant (44,913 acres)
- Indian Nations National Scenic And Wildlife Area (44,642 acres)
- Kenwood Indian Reservation (36,817 acres)
- Wister Wildlife Management Area & Waterfowl Refuge (Department of Wildlife Conservation; 35,425 acres)
- National Guard Camp Gruber (33,482 acres)
- Black Kettle National Grassland (USDA Forest Service; 30,688 acres)
- Greater Flint Hills (The Nature Conservancy; 28,570 acres)
- Oologah Lake (28,333 acres)
- Beaver River Wildlife Management Area (Department of Wildlife Conservation; 26,749 acres)
- Winding Stair Mountain National Recreation Area (26,617 acres)
- Eufaula Wildlife Management Area (Department of Wildlife Conservation; 23,829 acres)
- Copan Wildlife Management Area (Department of Wildlife Conservation; 23,456 acres)
- Keystone Wildlife Management Area (Department of Wildlife Conservation; 22,405 acres)
- Keystone Lake (22,342 acres)
- James Collins Wildlife Management Area (Department of Wildlife Conservation; 21,498 acres)
- Salt Plains National Wildlife Refuge (US Fish and Wildlife Service; 21,190 acres)
- Packsaddle Wildlife Management Area (Department of Wildlife Conservation; 20,513 acres)
- Hulah Wildlife Management Area (Department of Wildlife Conservation; 19,825 acres)
- ... and 1,546 more protected areas ...

Note: areas are listed based on name, ownership, and boundary information in the Protected Areas Database of the United States, which may include overlapping and duplicate areas. Only protected areas with at least 1 acre of overlap with this area are included in the list above.

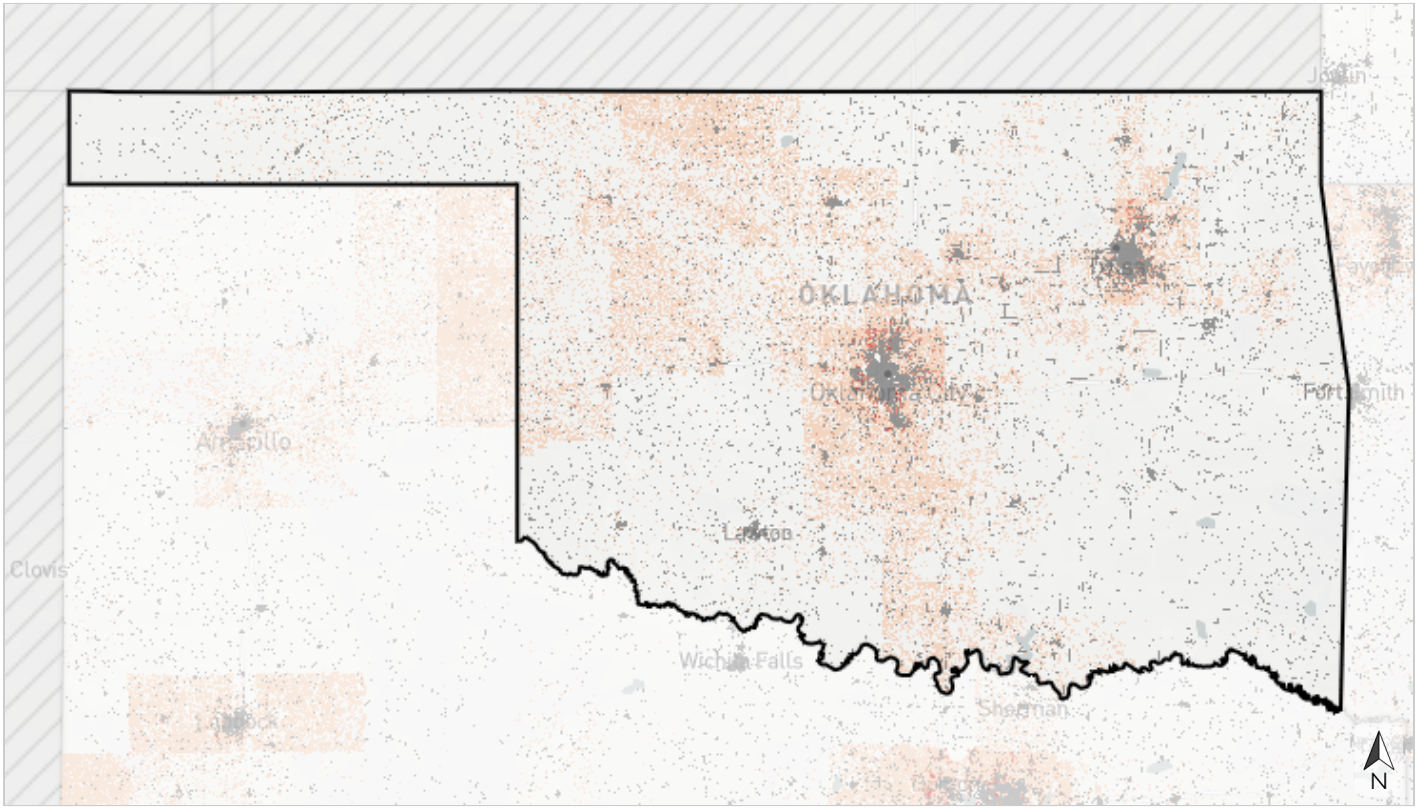
Sea Level Rise

Sea level rise unlikely to be a threat (inland counties).

Urban Growth

The FUTURES urban growth model predicts the likelihood that an area will urbanize at every decade from 2020 to 2100. Developed areas from the 2021 National Landcover Database serve as the baseline for current urban areas. The model simulates landscape change based on trends in population growth, local development suitability factors, and an urban patch-growing algorithm. It considers environmental drivers like distance to floodplain, slope, and available infrastructure, and even socio-economic status. The probability of urbanization for each area reflects how many times it urbanized out of 50 model runs.

To explore maps for additional time periods, click here.



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Probability of urbanization by 2060

- Urban in 2021
- Very high likelihood of urbanization (>50% probability)
- High likelihood of urbanization (25 - 50% probability)
- Moderate likelihood of urbanization (2 - 25% probability)
- Not likely to urbanize

Table 28: Extent of projected urbanization by decade within Oklahoma. Values from FUTURES model projections for the contiguous United States developed by the Center for Geospatial Analytics, NC State University. 2060 corresponds to the SECAS goal: a 10% or greater improvement in the health, function, and connectivity of Southeastern ecosystems by 2060.

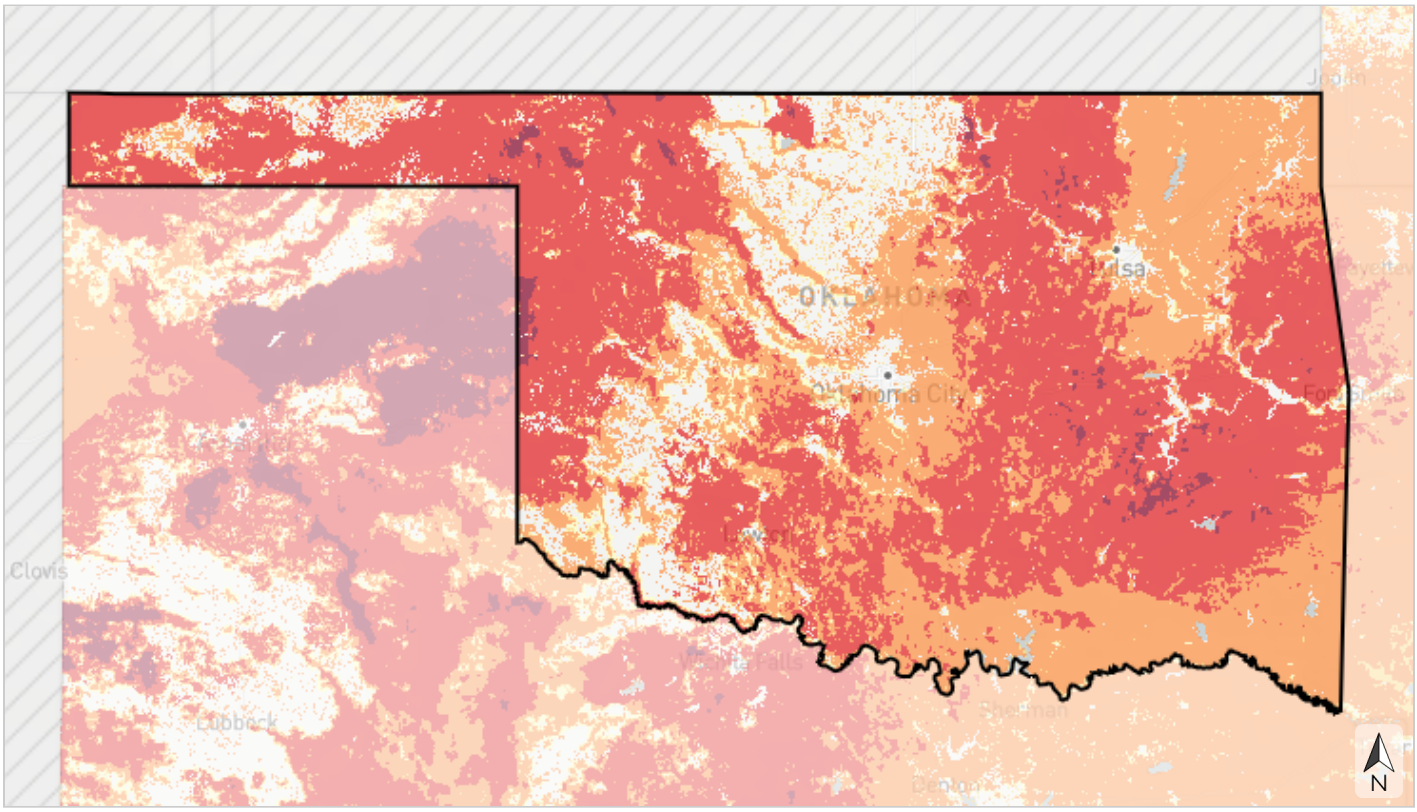
Decade	Acres	Percent of Area
Urban in 2021	2,556,232	5.7%
2030 projected extent	2,667,747	6.0%
2040 projected extent	2,736,499	6.1%
2050 projected extent	2,802,211	6.3%
2060 projected extent	2,871,394	6.4%
2070 projected extent	2,933,138	6.6%
2080 projected extent	2,986,734	6.7%
2090 projected extent	3,029,963	6.8%
2100 projected extent	3,063,925	6.8%
Not projected to urbanize by 2100	*41,671,061*	*93.2%*
Total area	**44,734,986**	**100%**

5.7% of this area is already urban in 2021, and an additional 15.0% has at least a moderate probability of urbanizing by 2060.

By 2060, the size of the urban footprint is projected to increase **12%** over 2021 levels.

Wildfire Likelihood

Wildfire likelihood data originate from the Wildfire Risk to Communities project developed by the U.S. Forest Service. This layer depicts the probability of wildfire burning in a specific location in any given year. Annual burn probabilities in the United States range from 0-14%, but do not exceed 8% in the Southeast. Wildfire likelihood is based on fire behavior modeling across thousands of simulations of possible fire seasons. In each simulation, factors contributing to the probability of a fire occurring (such as weather, topography, and ignitions) vary based on patterns derived from observations in recent decades. Wildfire likelihood is not predictive and does not reflect any forecasted future weather or fire danger conditions. It also does not say anything about the intensity of fire if it occurs. To explore additional wildfire risk information, please see the Wildfire Risk to Communities website.



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Wildfire likelihood (annual burn probability)

- High
- Moderate-high
- Moderate
- Low-moderate
- Low
- Not predicted to experience wildfire

Table 29: Area in each wildfire probability category within Oklahoma. Values from the Wildire Risk To Communities project developed by the USDA Forest Service.

Wildfire likelihood (annual burn probability)	Acres	Percent of Area
Not predicted to experience wildfire (0% probability)	5,435,075	12.1%
Low (>0 - 0.01% probability)	1,447,503	3.2%
Low-moderate (>0.01 - 0.02154% probability)	511,973	1.1%
Low-moderate (>0.02154 - 0.04642% probability)	806,130	1.8%
Moderate (>0.04642 - 0.1% probability)	1,711,246	3.8%
Moderate (>0.1 - 0.21544% probability)	4,822,048	10.8%
Moderate (>0.21544 - 0.46416% probability)	8,760,228	19.6%
Moderate-high (>0.46416 - 1% probability)	12,690,472	28.4%
Moderate-high (>1 - 2.15443% probability)	8,080,914	18.1%
High (>2.15443 - 4.64159% probability)	469,295	1.0%
High (>4.64159% probability)	0	0%
No wildfire risk data available	*102*	*<0.1%*
Total area	**44,734,986**	**100%**